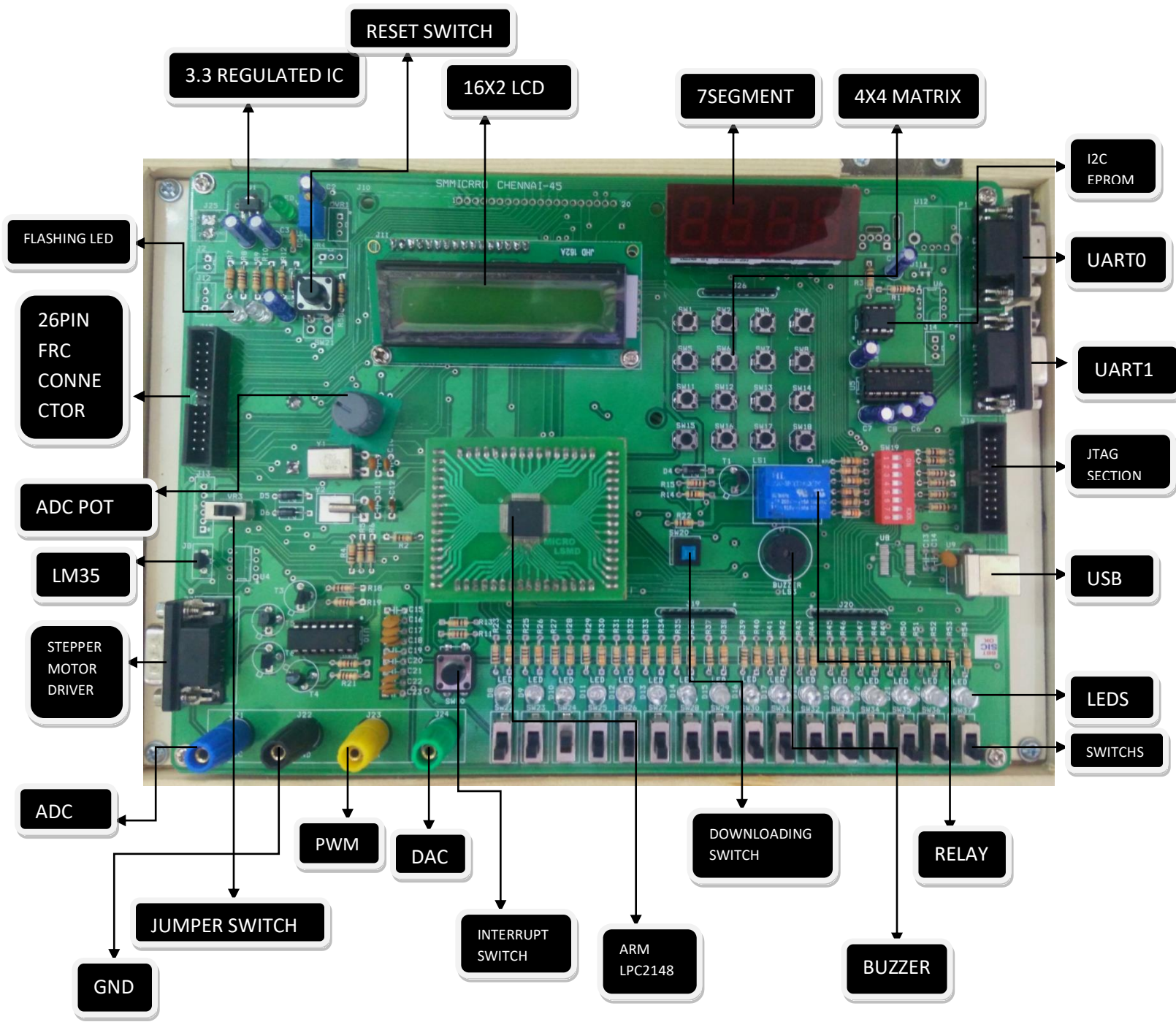


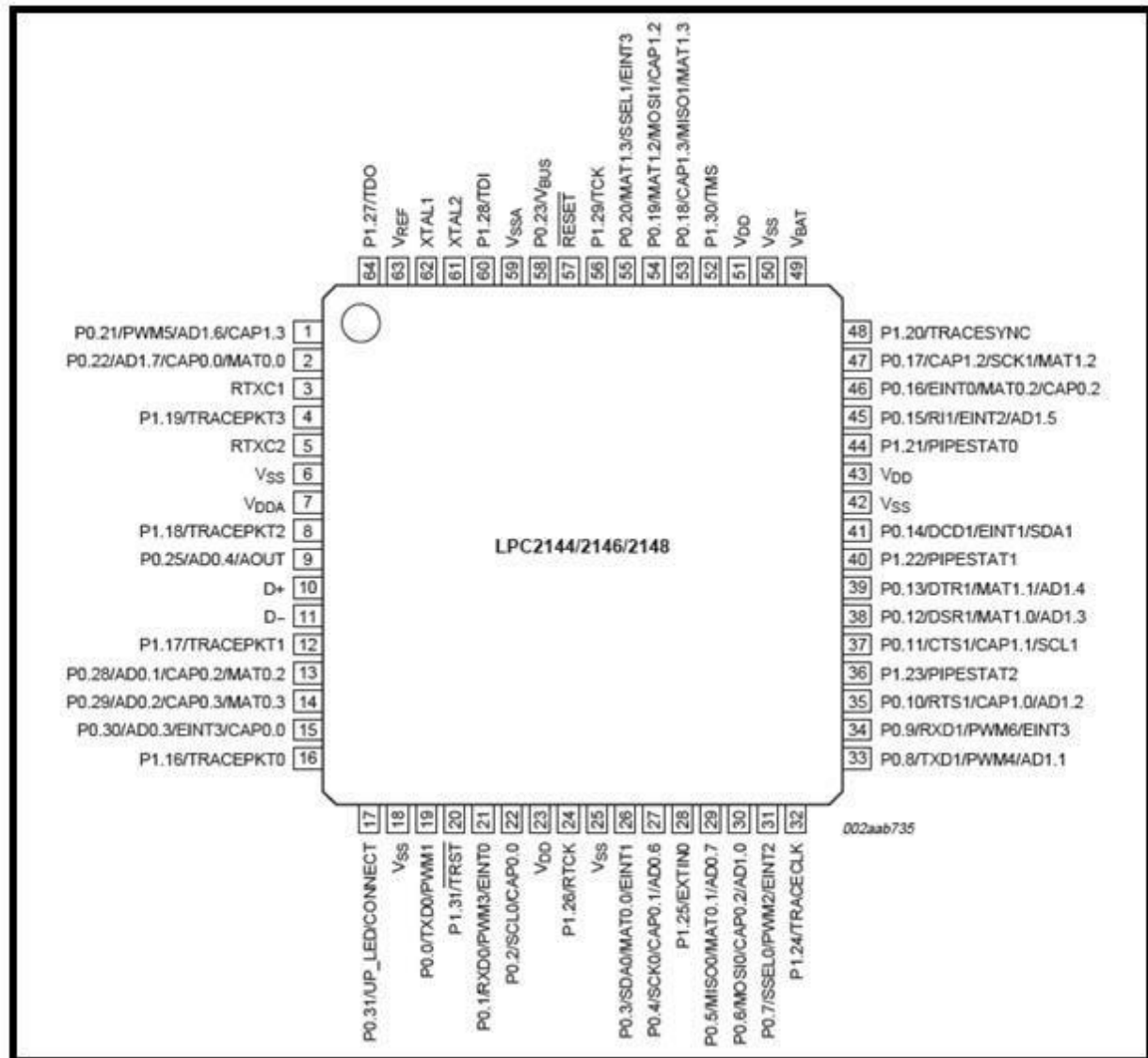
ARM LPC2148 TRAINER KIT



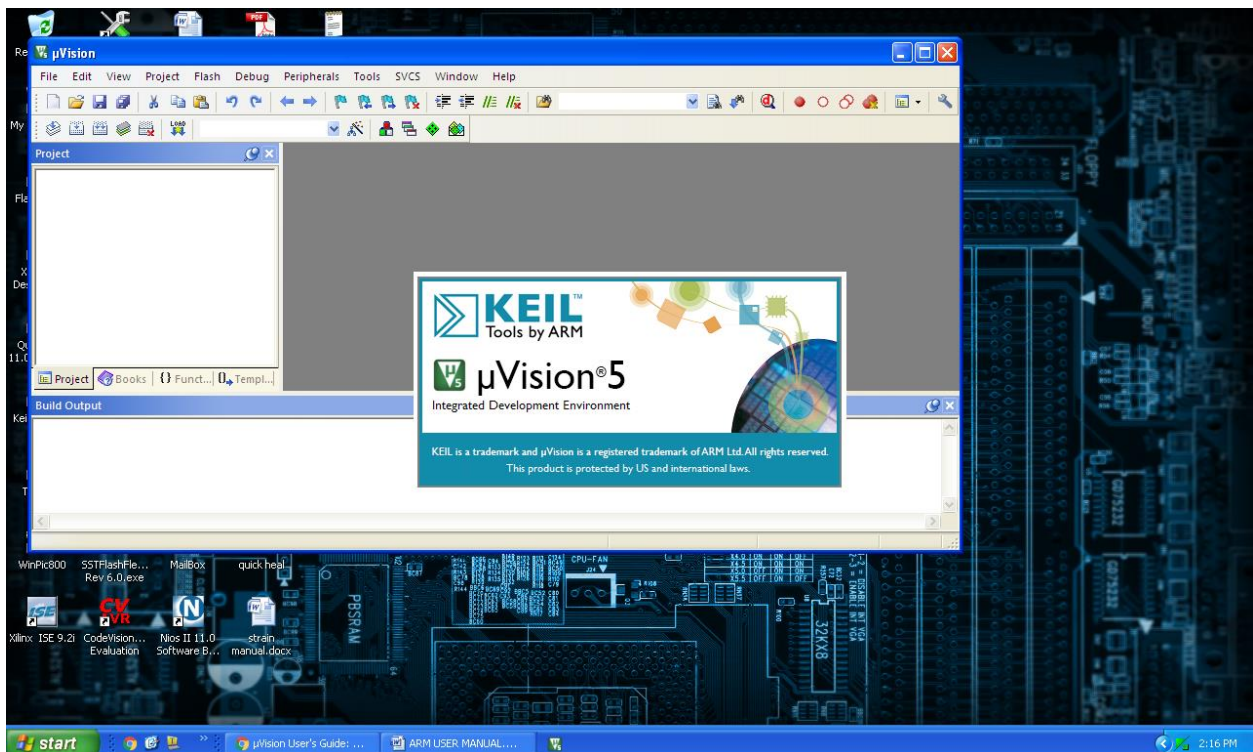
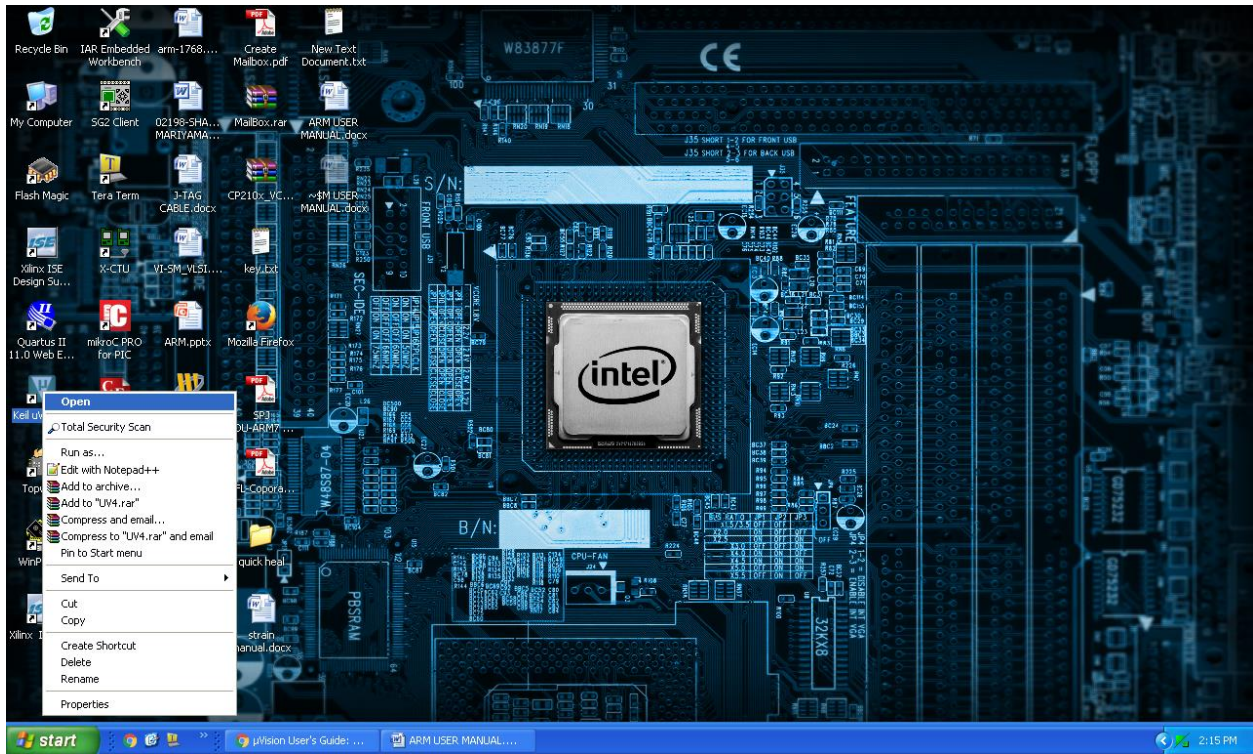
LPC2148 PROCESSOR FEATURES

- 16-bit/32-bit **ARM7 microcontroller** in a tiny LQFP64 package.
- 8 kB to 40 kB of on-chip static RAM and 32 kB to 512 kB of on-chip flash memory. 128-bit wide interface/accelerator enables high-speed 60 MHz operation.
- In-System Programming/In-Application Programming (ISP/IAP) via on-chip boot loader software. Single flash sector/full chip erase in 400 ms and programming of 256 bytes in 1 ms.
- USB 2.0 Full-speed compliant device controller with 2 kB of endpoint RAM. In addition, the LPC2146/48 provides 8 kB of on-chip RAM accessible to USB by DMA.
- One or two (LPC2141/42 vs. LPC2144/46/48) 10-bit ADCs provide a total of 6/14 analog inputs, with conversion times as low as 2.44 μ s per channel.
- Single 10-bit DAC provides variable analog output (LPC2142/44/46/48 only).
- Two 32-bit timers/external event counters (with four capture and four compare channels each), PWM unit (six outputs) and watchdog.
- Low power Real-Time Clock (RTC) with independent power and 32 kHz clock input. Multiple serial interfaces including two UARTs (16C550), two Fast I2C-bus (400 Kbit/s), SPI and SSP with buffering and variable data length capabilities.
- Vectored Interrupt Controller (VIC) with configurable priorities and vector addresses.
- Up to 45 of 5 V tolerant fast general purpose I/O pins in a tiny LQFP64 package.
- Up to 21 external interrupt pins available.
- 60MHz maximum CPU clock available from programmable on-chip PLL with settling time of 100 μ s.
- On-chip integrated oscillator operates with an external crystal from 1 MHz to 25 MHz
- Power saving modes include Idle and Power-down.
- Individual enable/disable of peripheral functions as well as peripheral clock scaling for additional power optimization.
- Processor wake-up from Power-down mode via external interrupt or BOD.
- Single power supply chip with POR and BOD circuits:
- CPU operating voltage range of 3.0 V to 3.6 V (3.3 V $\pm$ 10 %) with 5 V tolerant I/O pads.

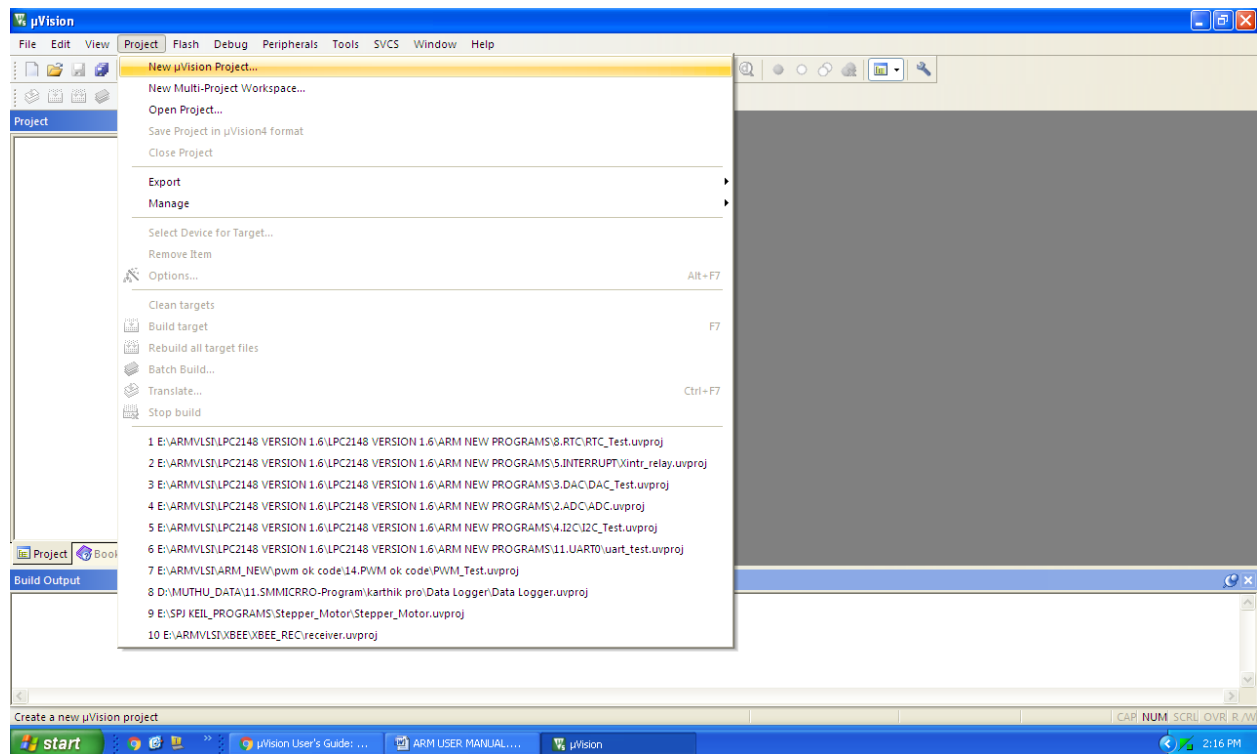
LPC2148 PIN DIAGRAM:



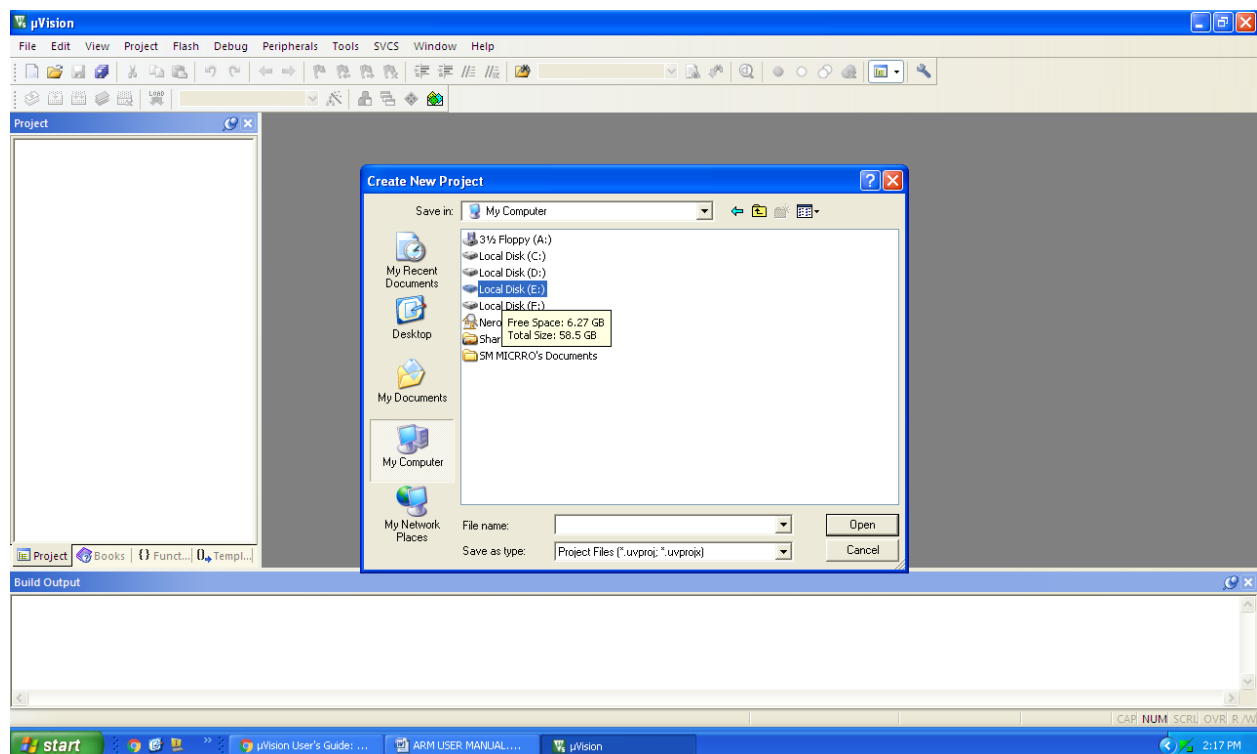
Step1: Open keil uvision.



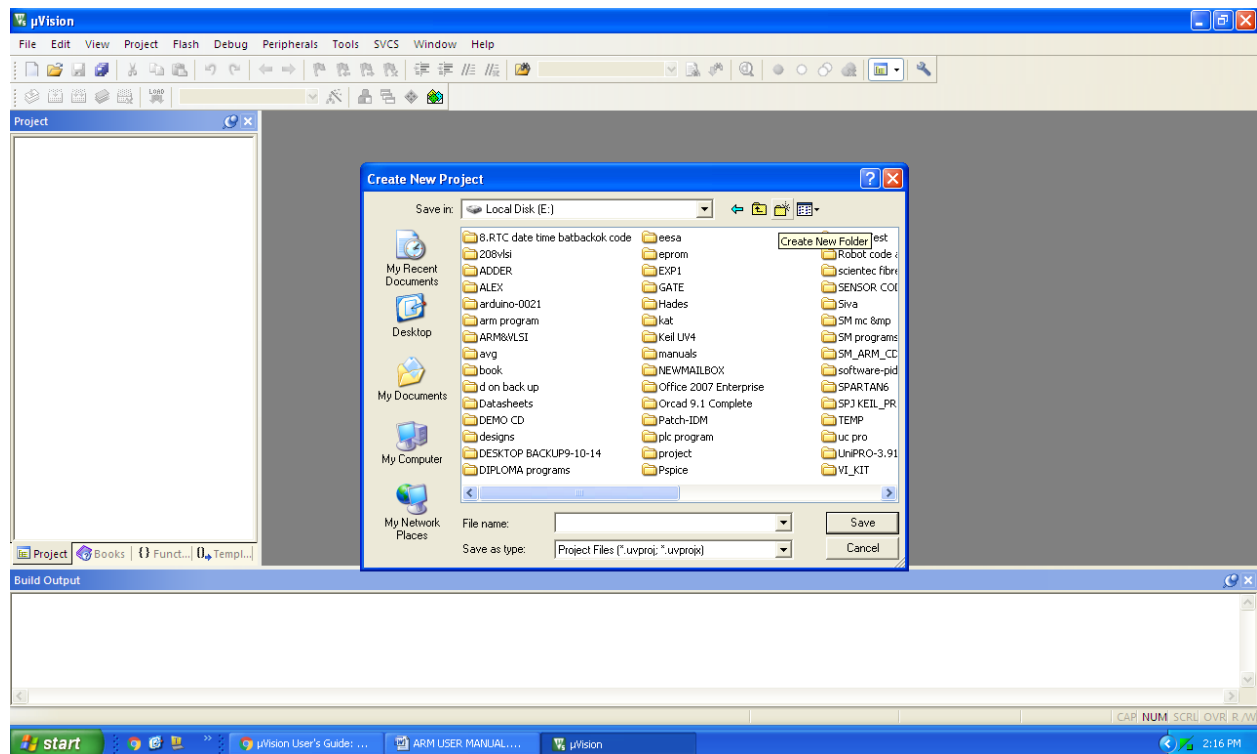
Step2: Project→New uvision project.



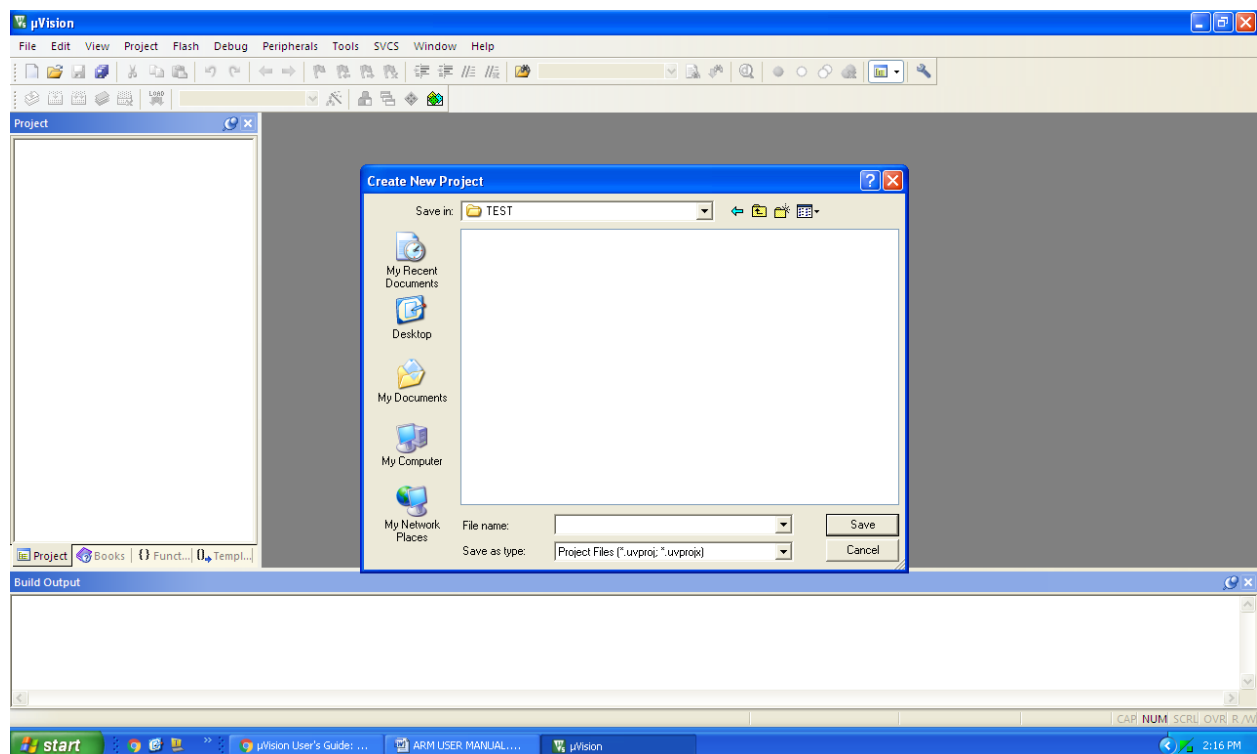
Step3: Select location where your project will save.



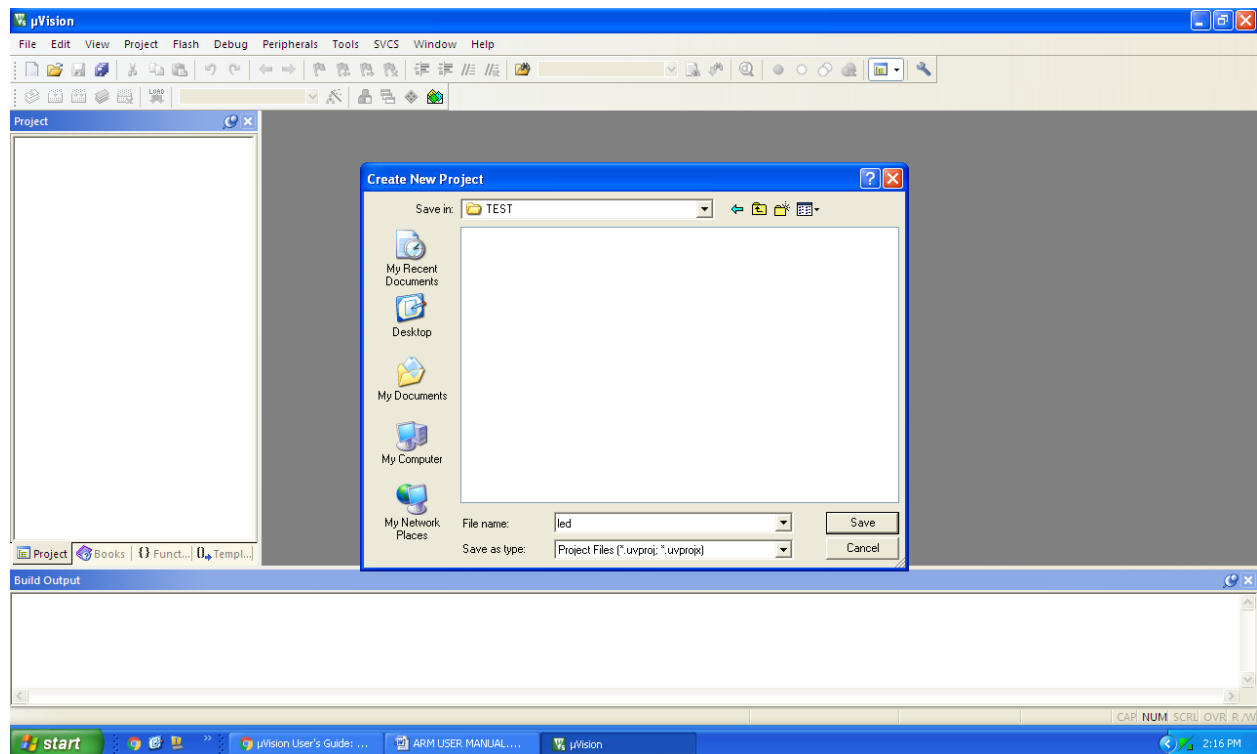
Step4: To create new folder and open the folder.



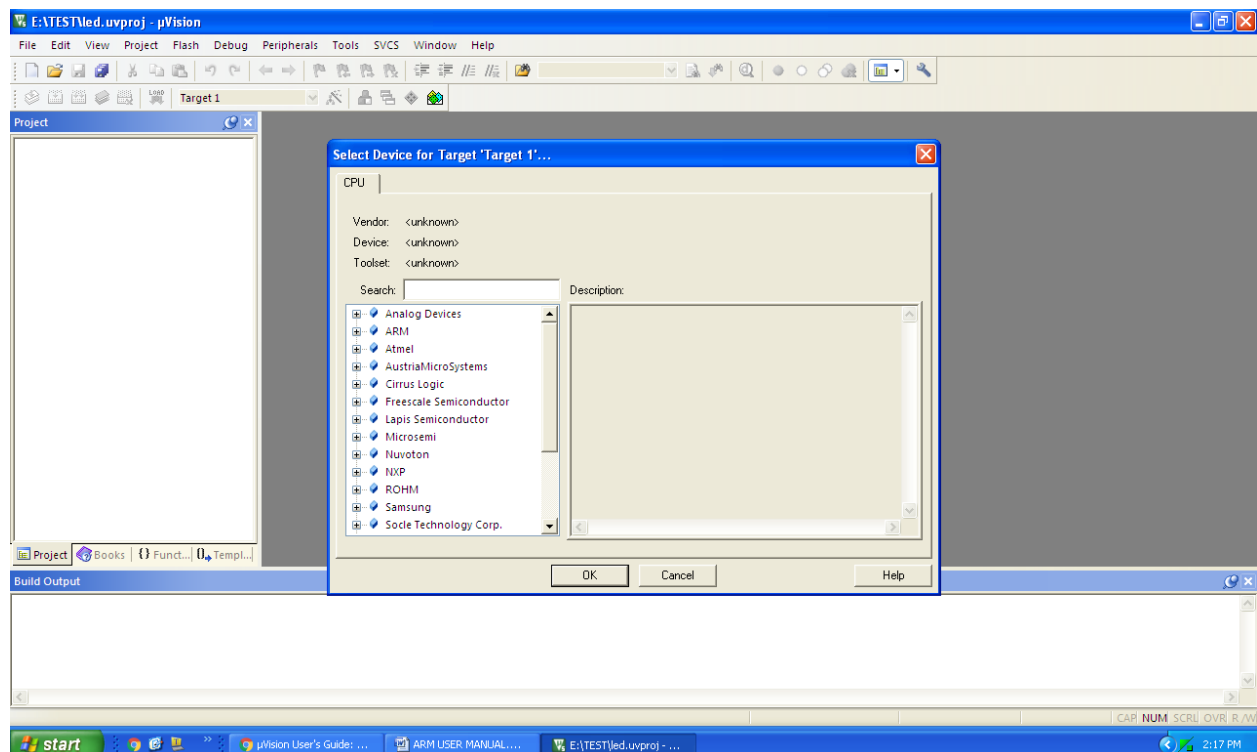
Step5: Here you give the project name.



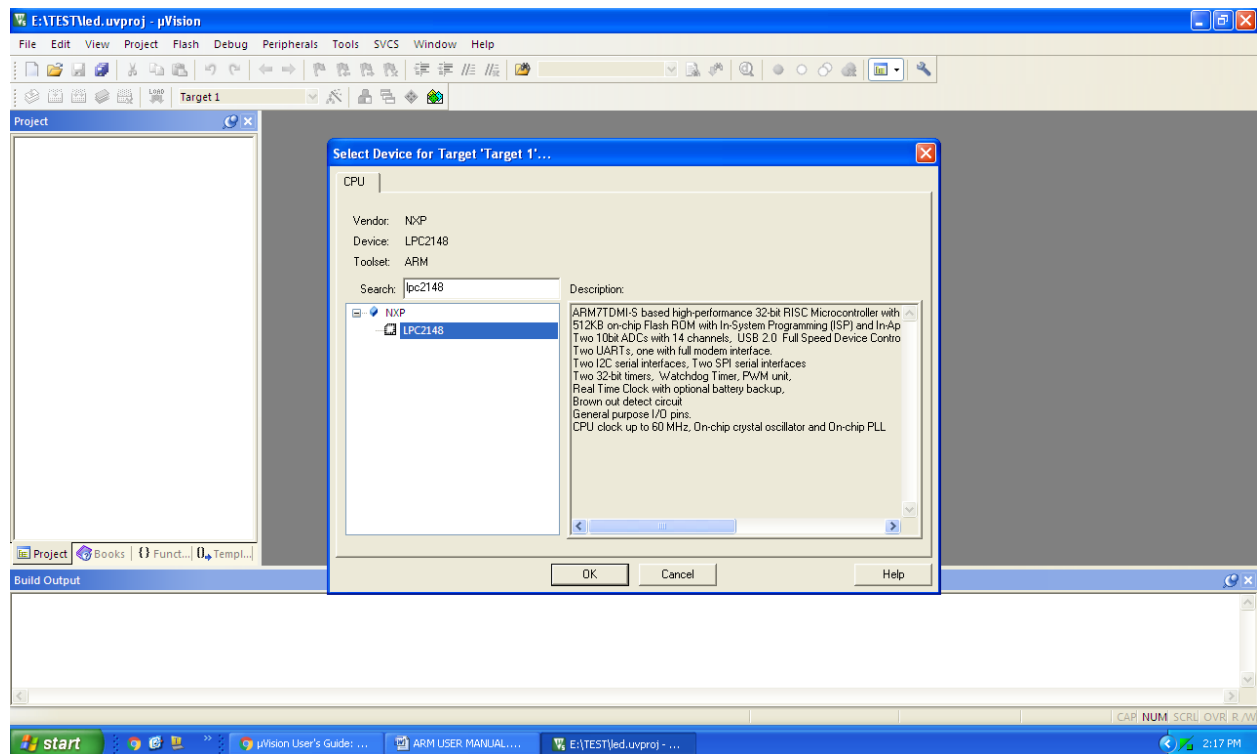
Step6: Give the project name and click save button.



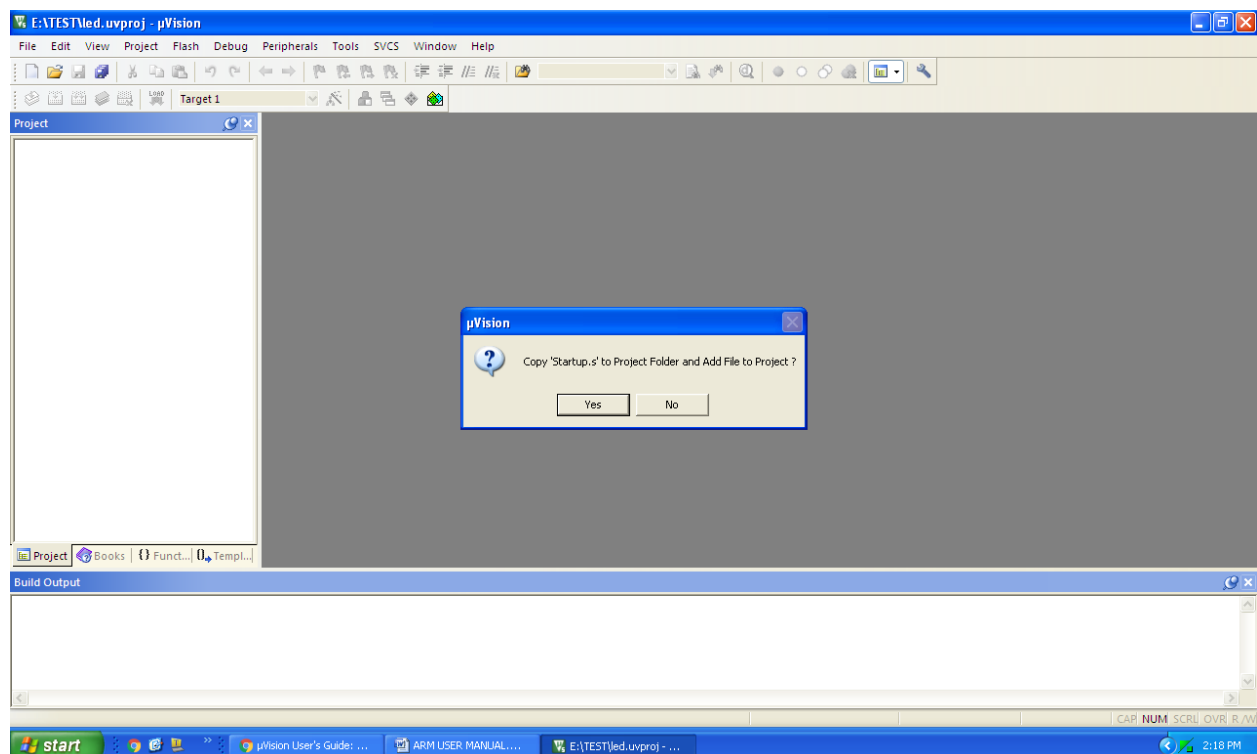
Step7: Here select the IC name click NXP→LPC2148. Otherwise type lpc2148 in search box.



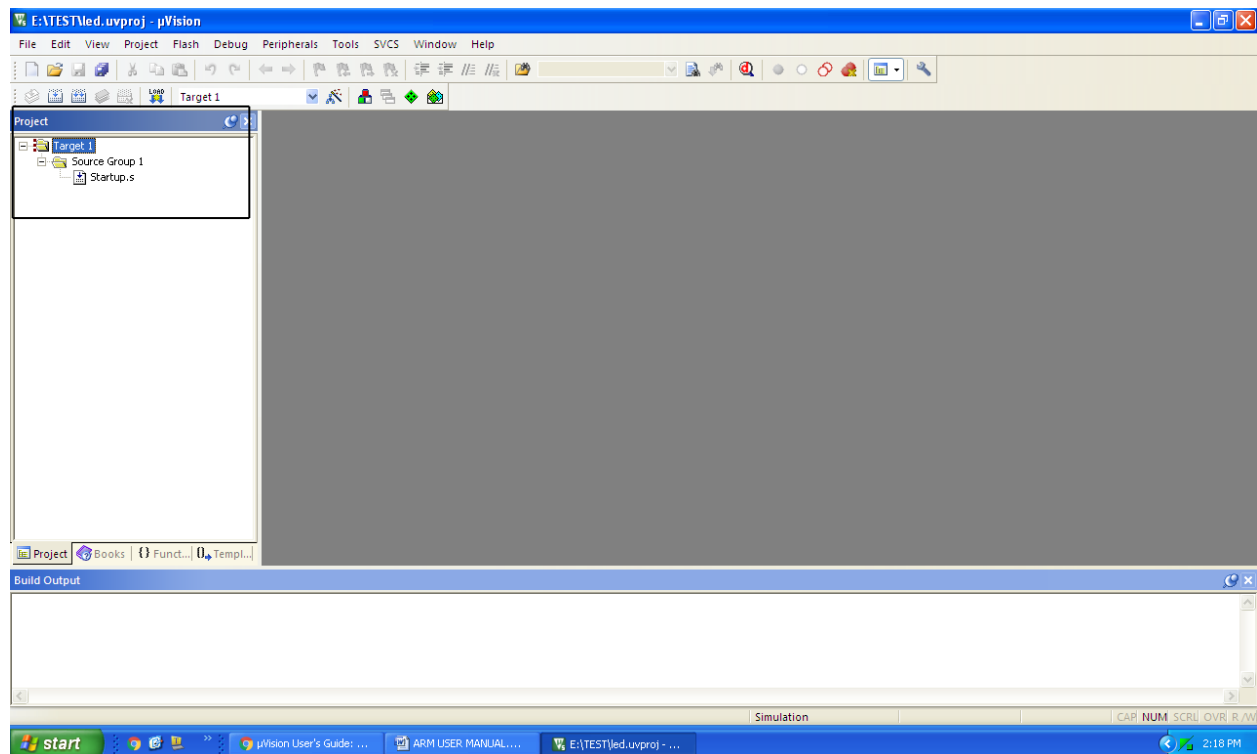
Step8: Select and click ok button.



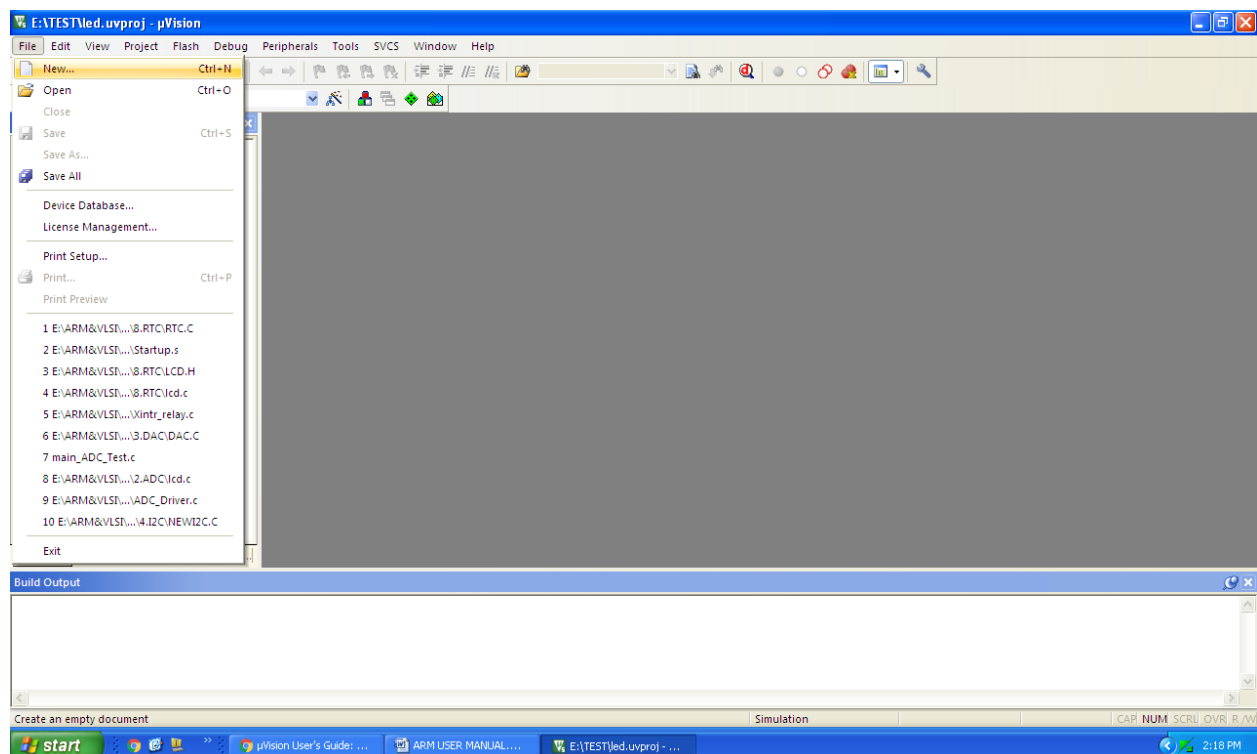
Step9: Here select yes button.



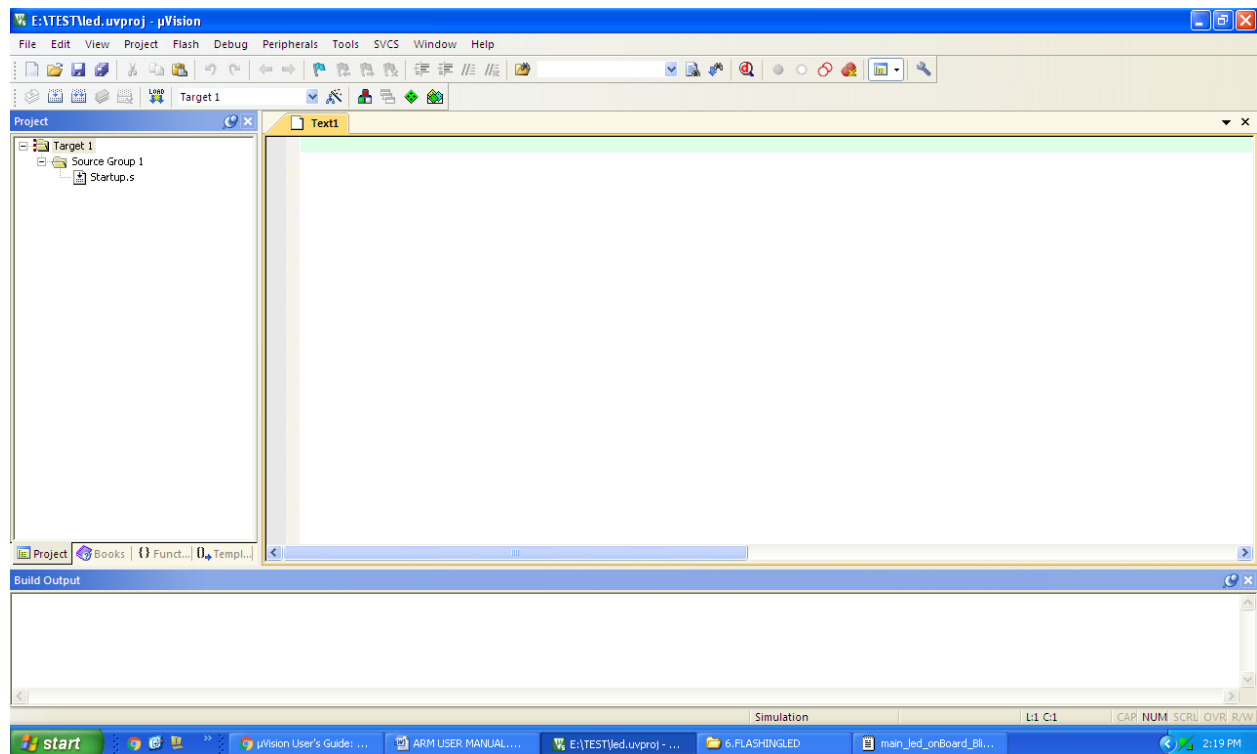
Step10: Left hand side our target will create.



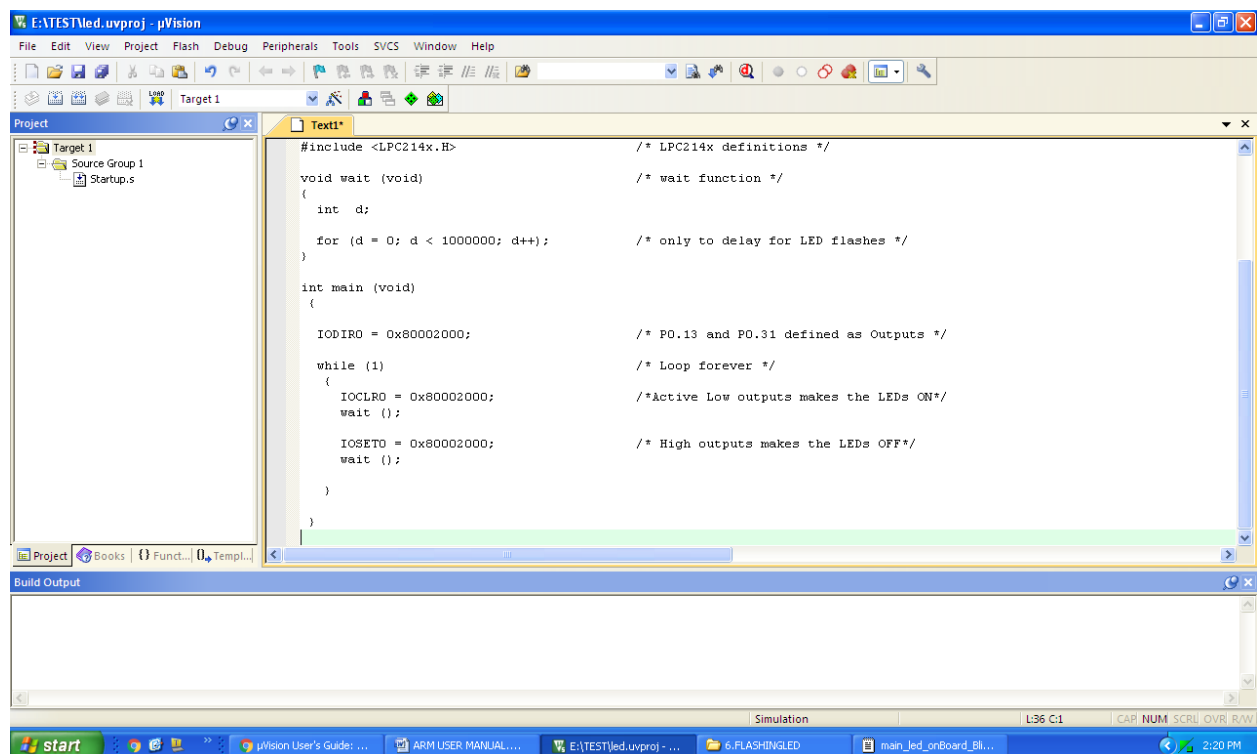
Step11: Next go to File→New.



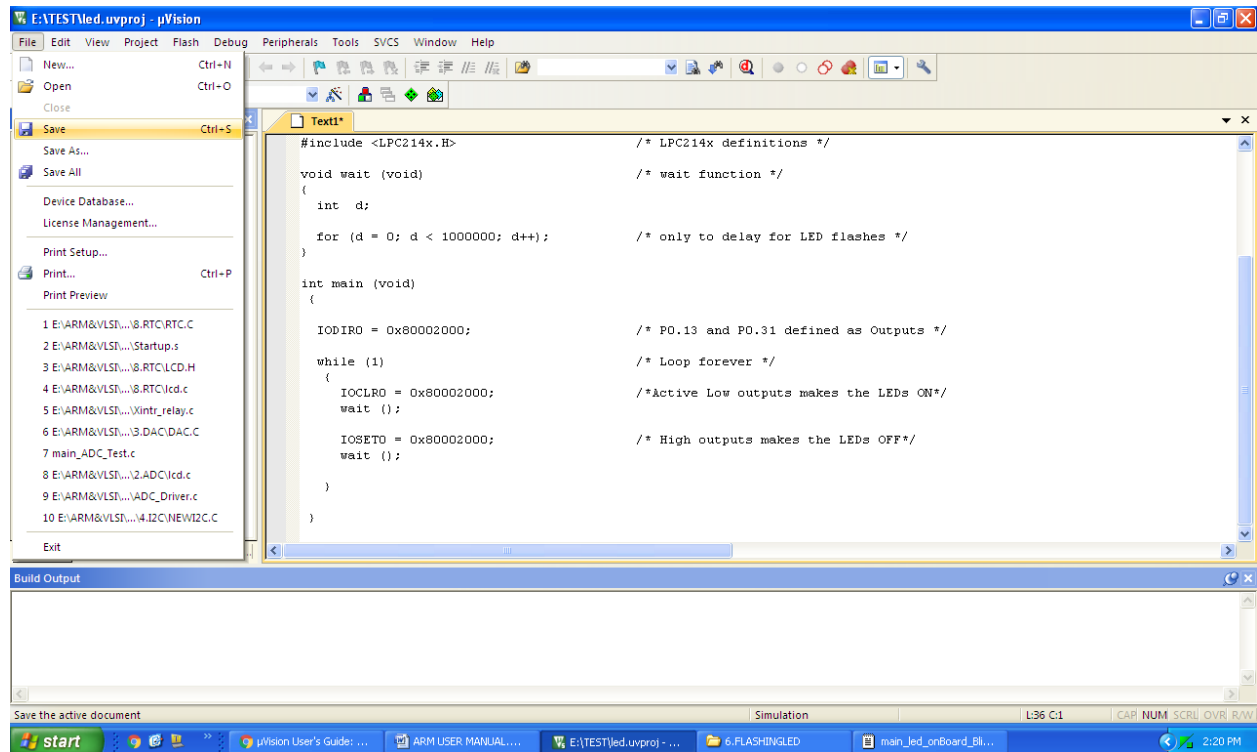
Step12: New workspace will open.



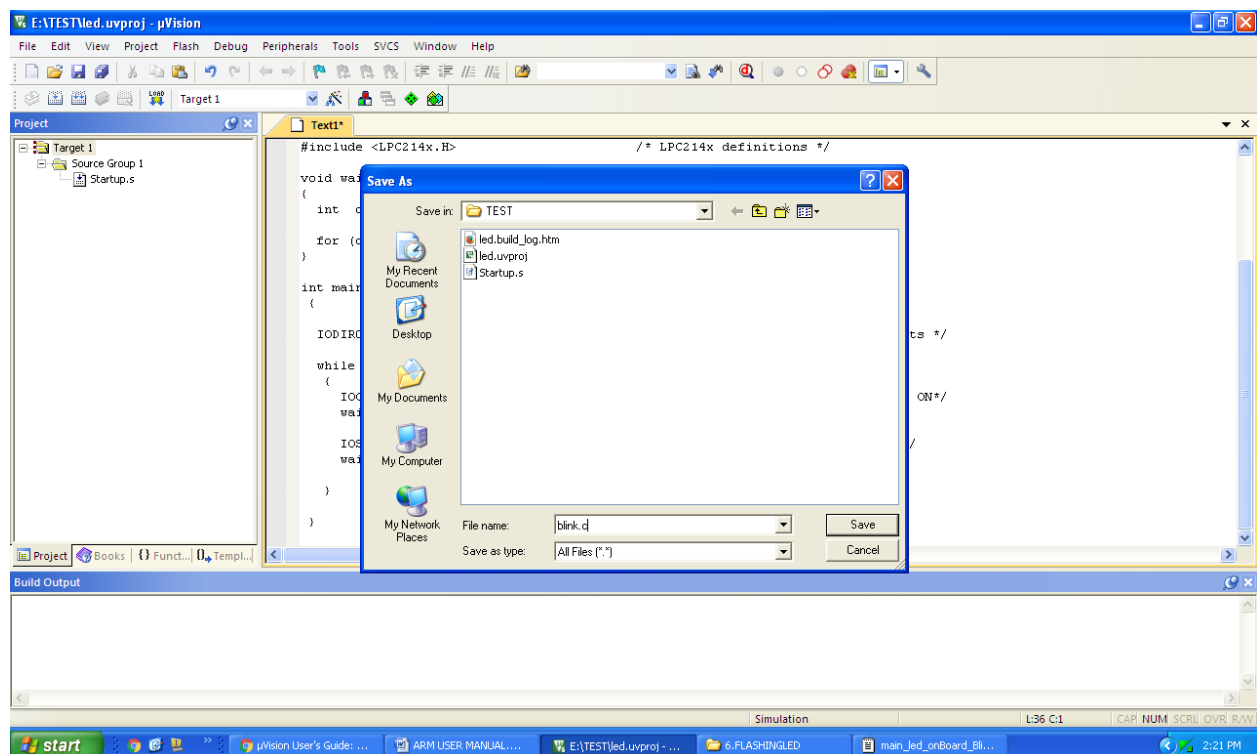
Step13: To copy and paste blinkinglead.c



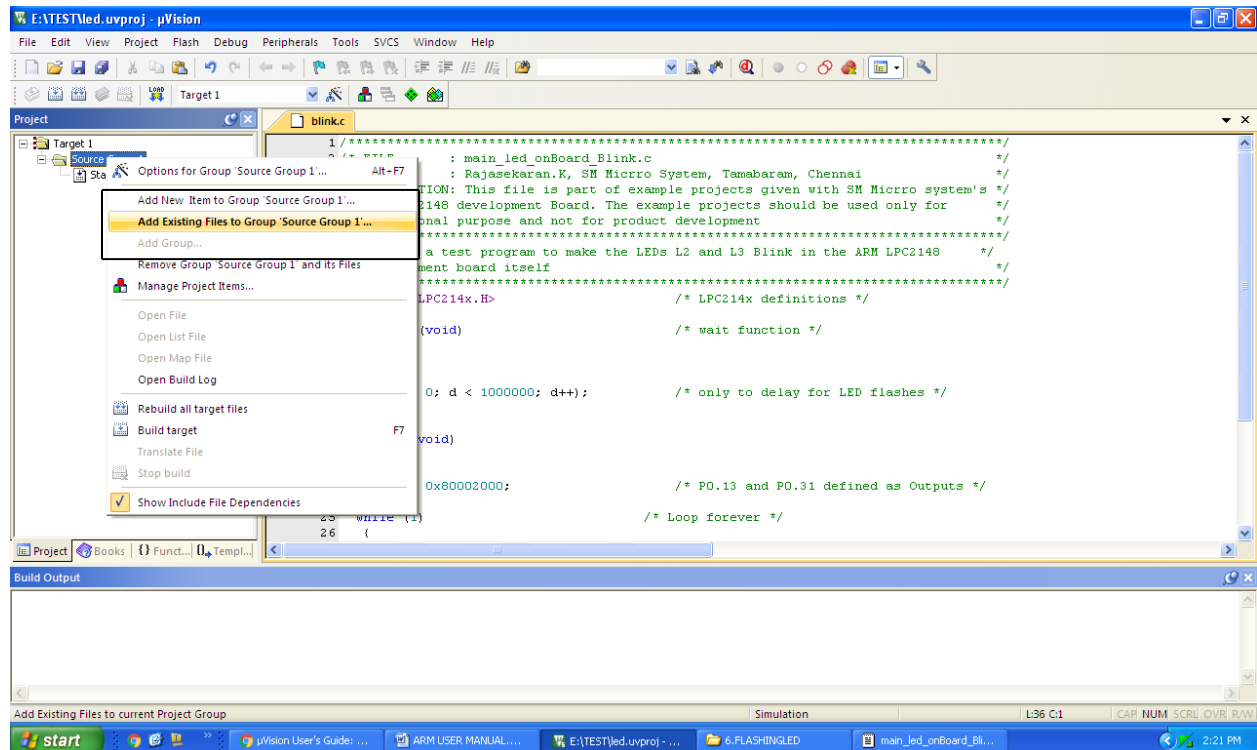
Step14: Next save the file like led.c.



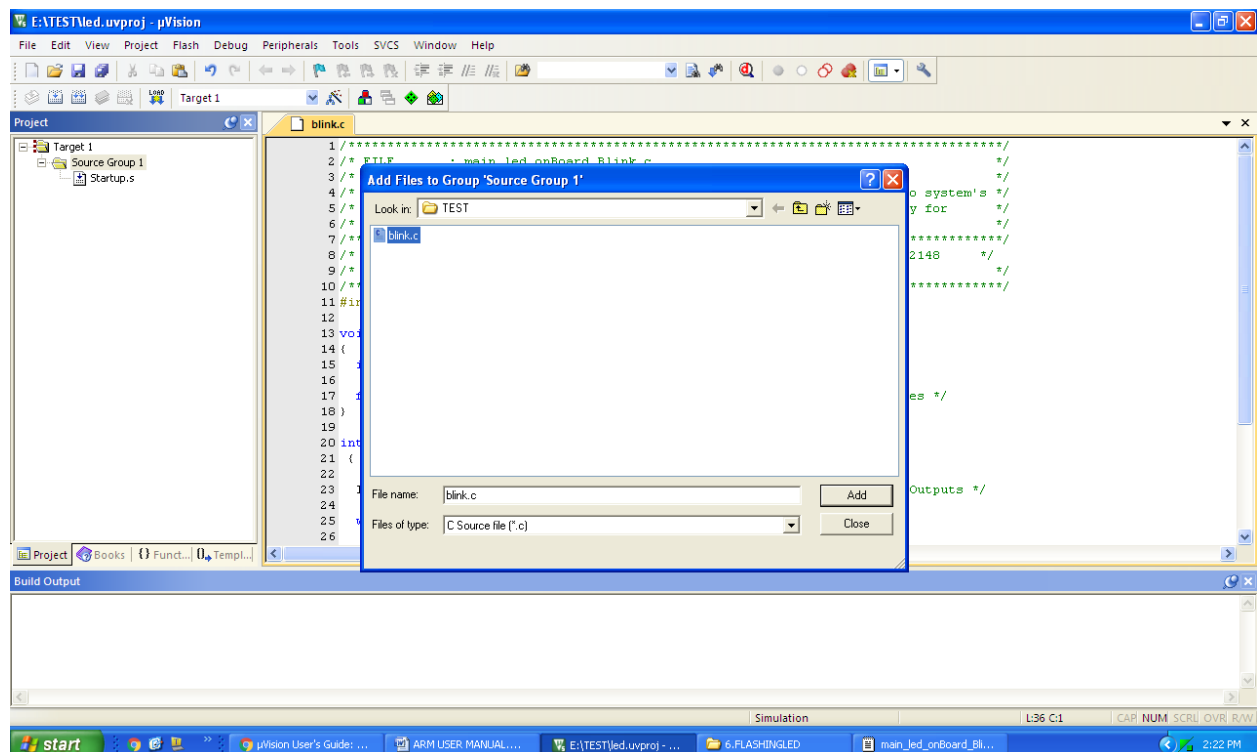
Step15: Must you have to save the file in .c extension.



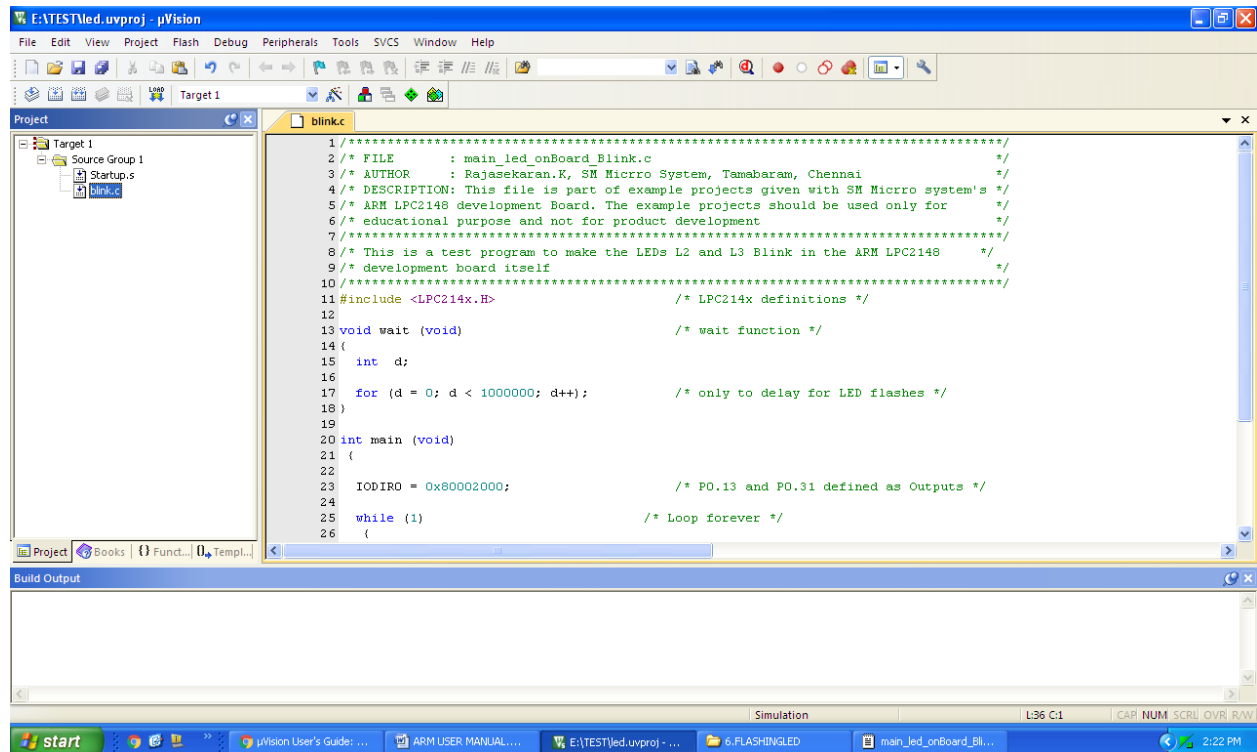
Step16: Before compile the project you must add the file in target. Right click on source group→add existing files to ‘source group1...’



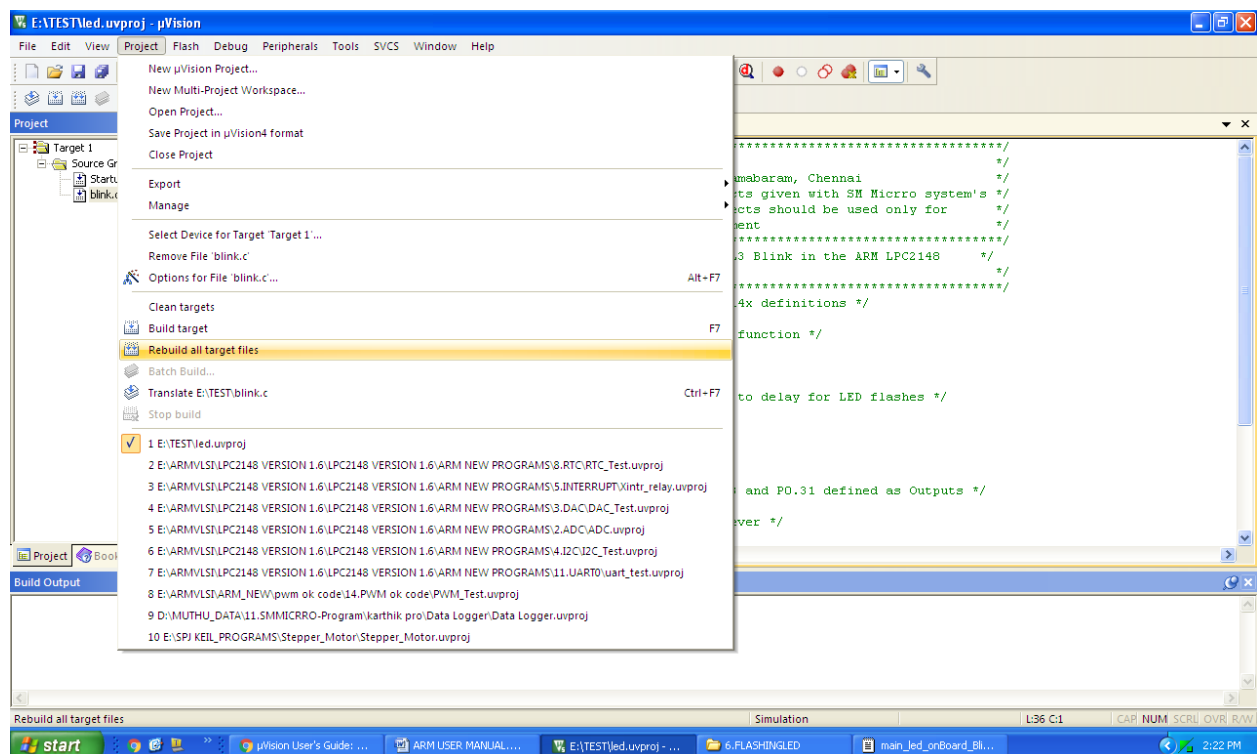
Step17: Select the file and click add button.



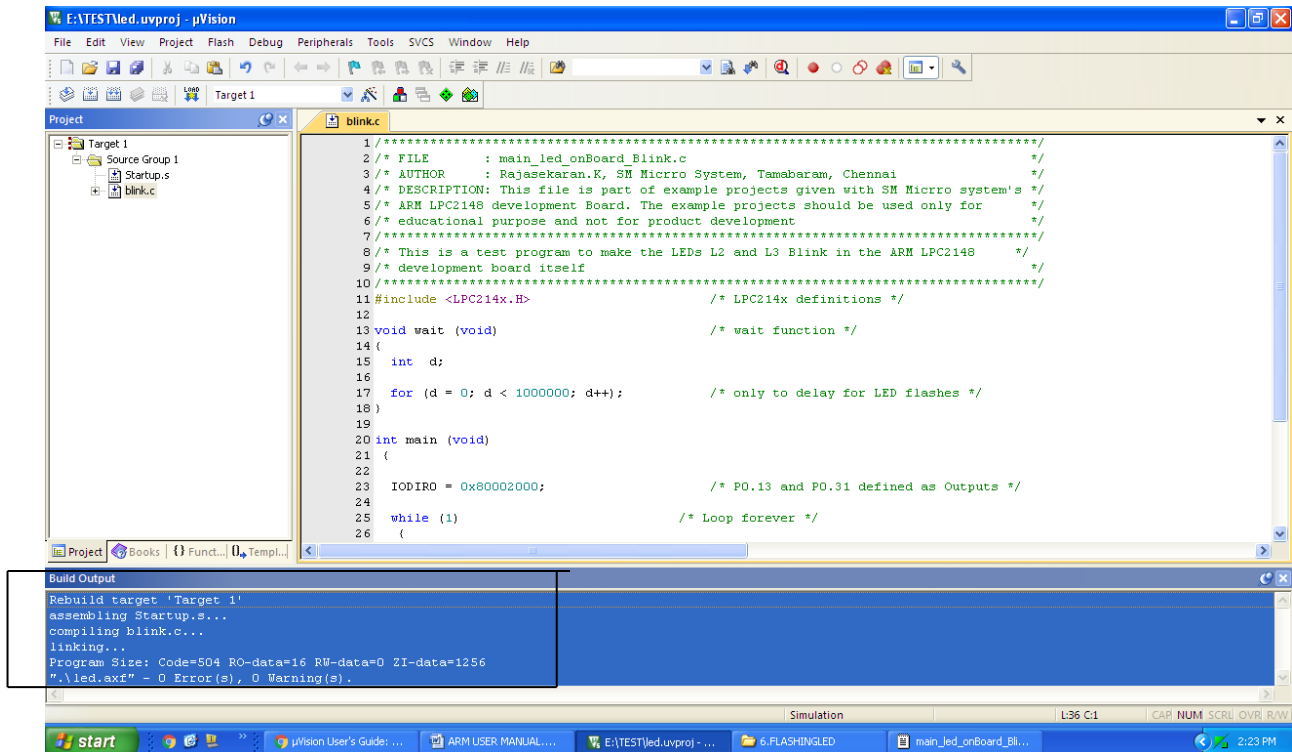
Step18: File will be add to our target.



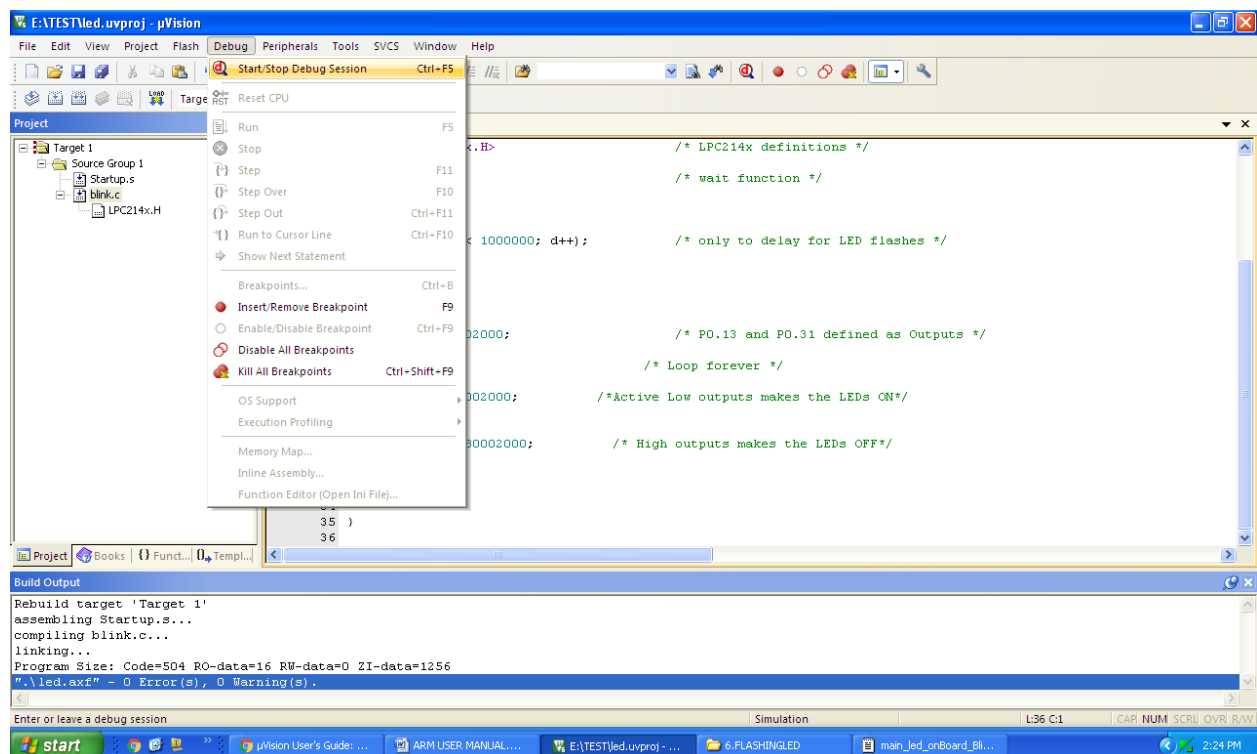
Step19: Go to project→rebuild all target file.



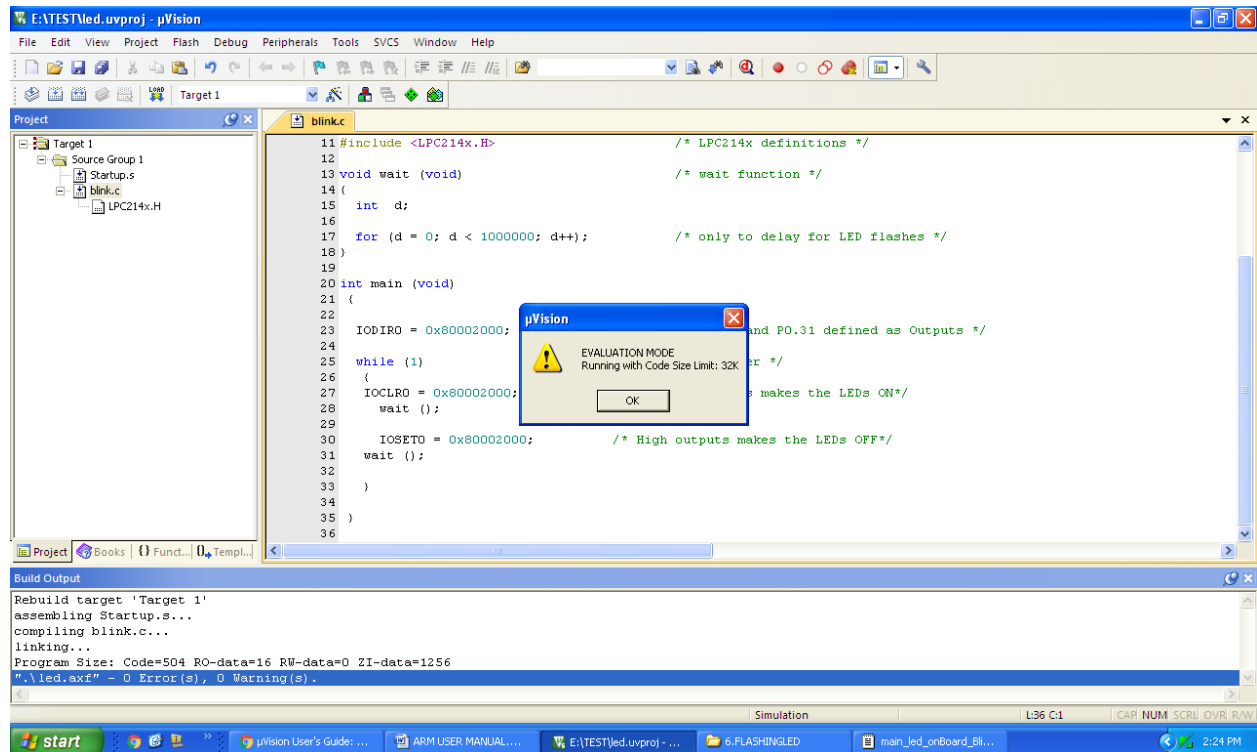
Step20: The program will compiled and build output



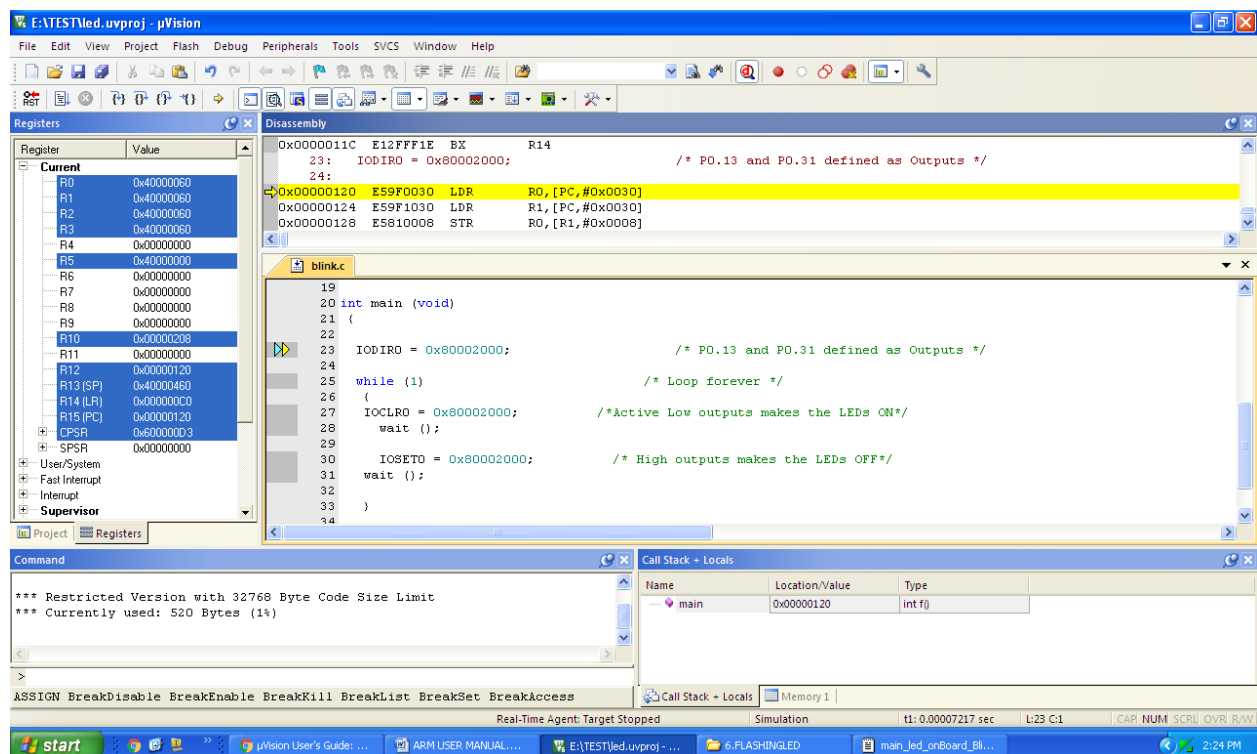
Step21: simulate our project. Debug→Start/Stop Debug Session.



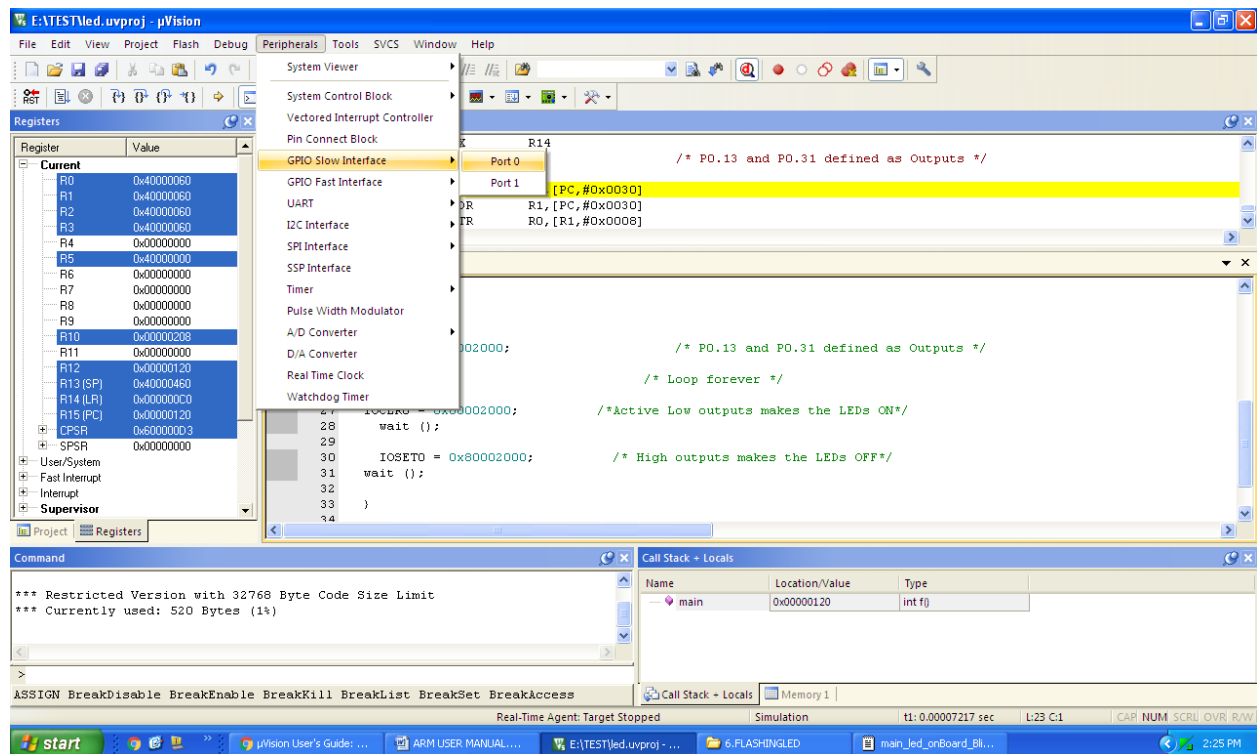
Step22: Click → Ok button.



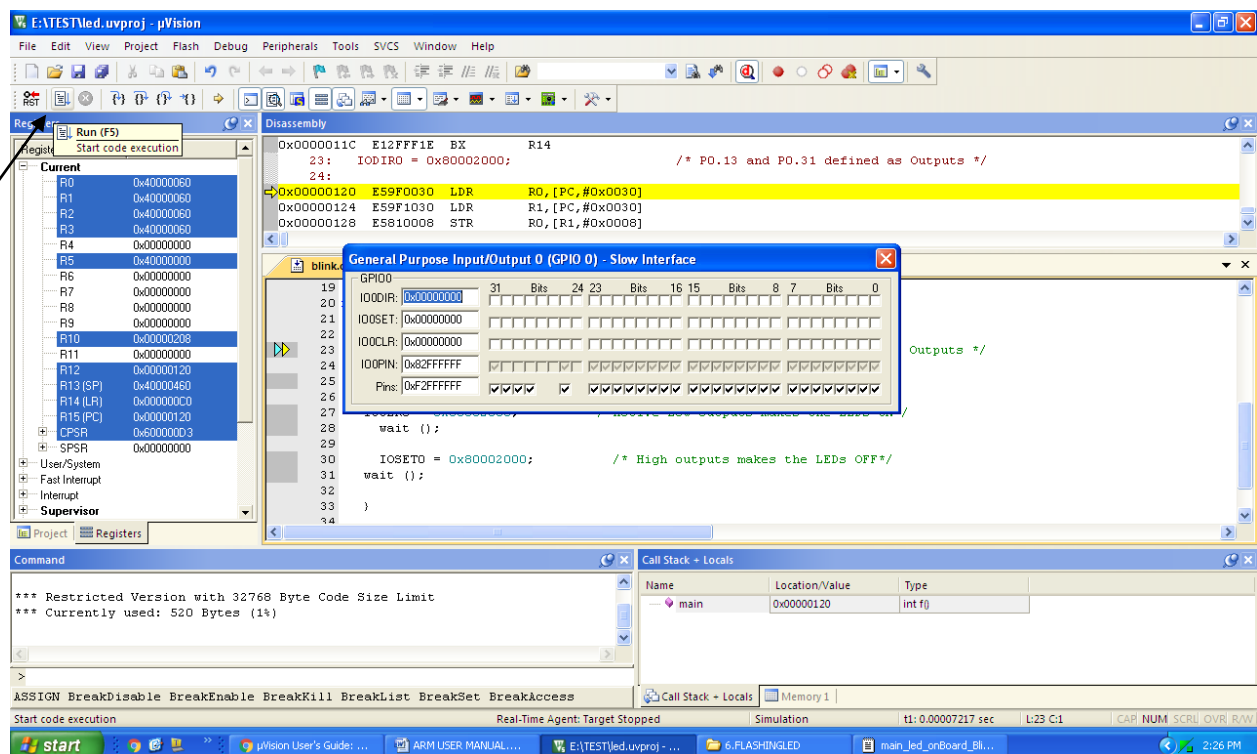
Step23: Simulation window was opened.



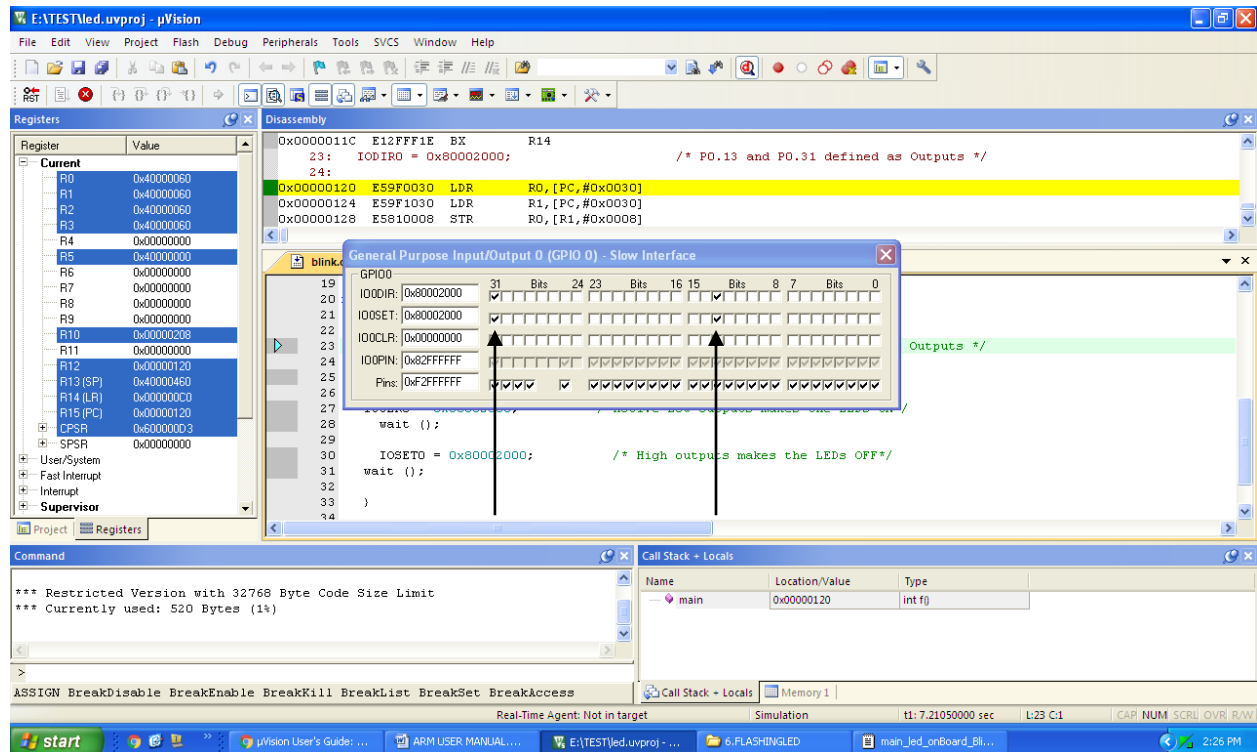
Step24: Peripherals → GPIO Slow Interface → Port0.



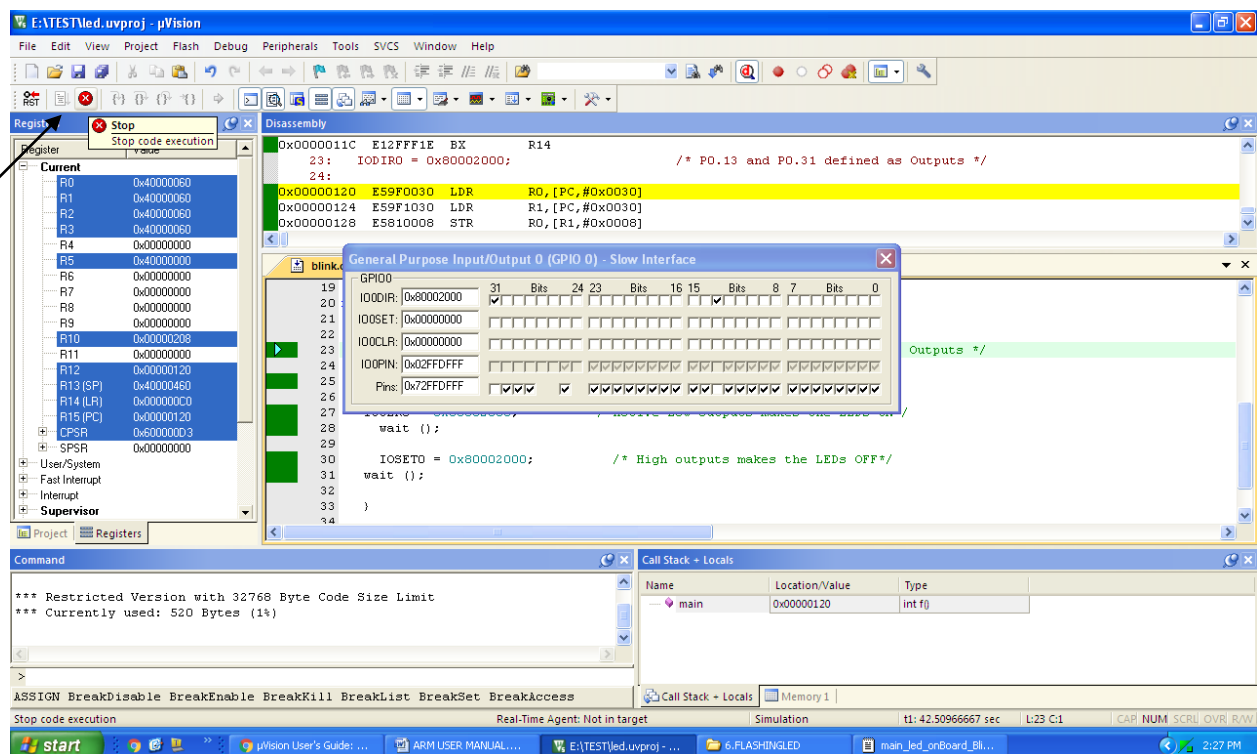
Step25: Port0 I/O pins open. And click the Run (F5).



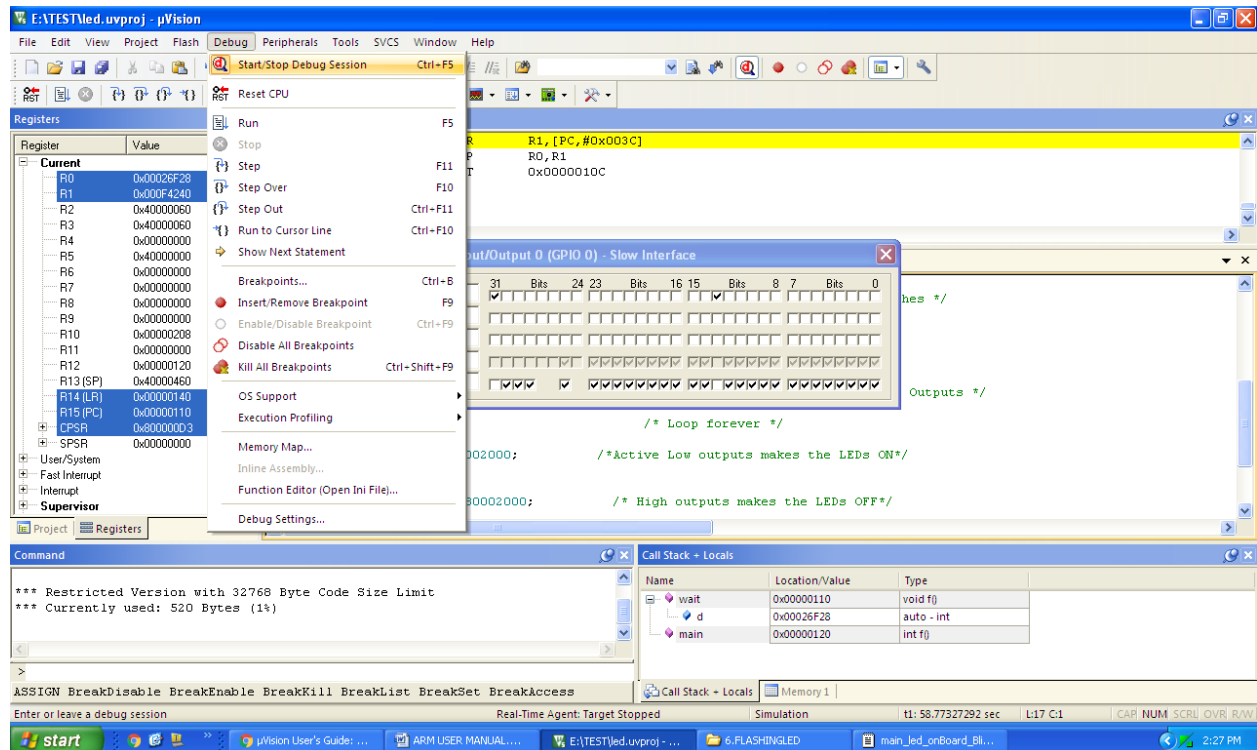
Step26: After click the run the program executed and 2 pins are blinked.



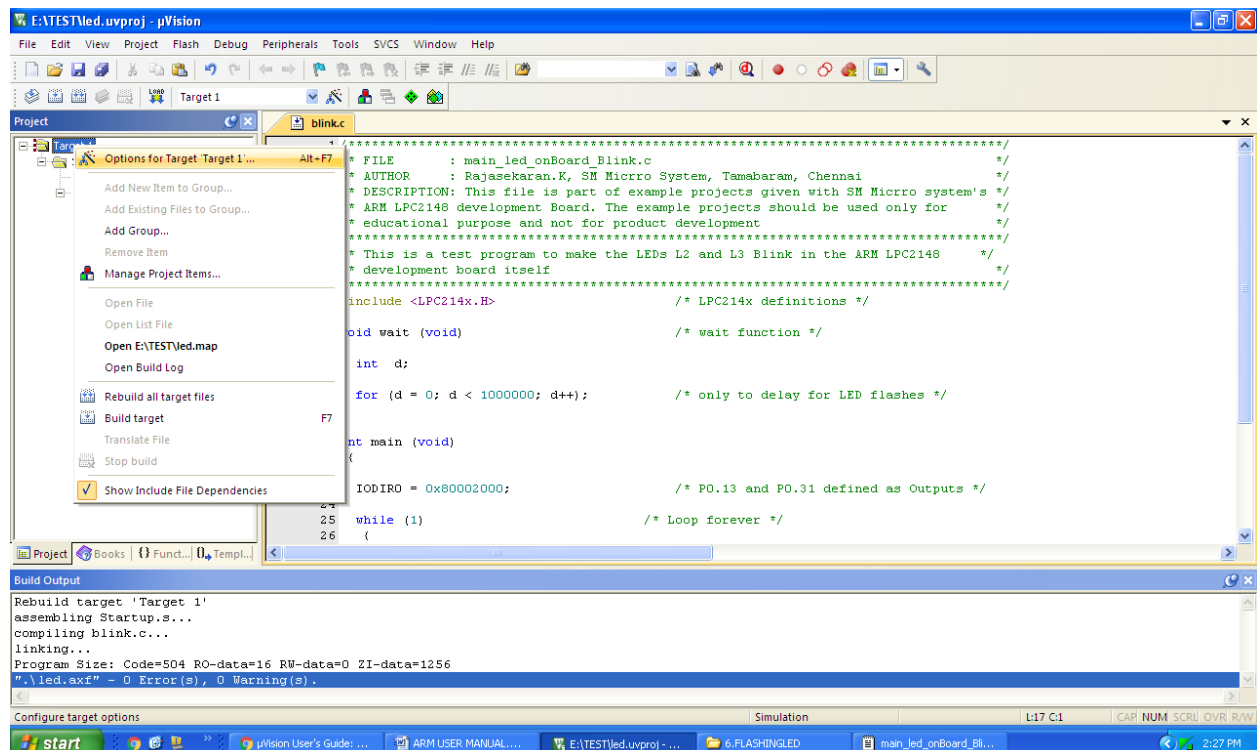
Step27: After seeing the outputs click the stop button.



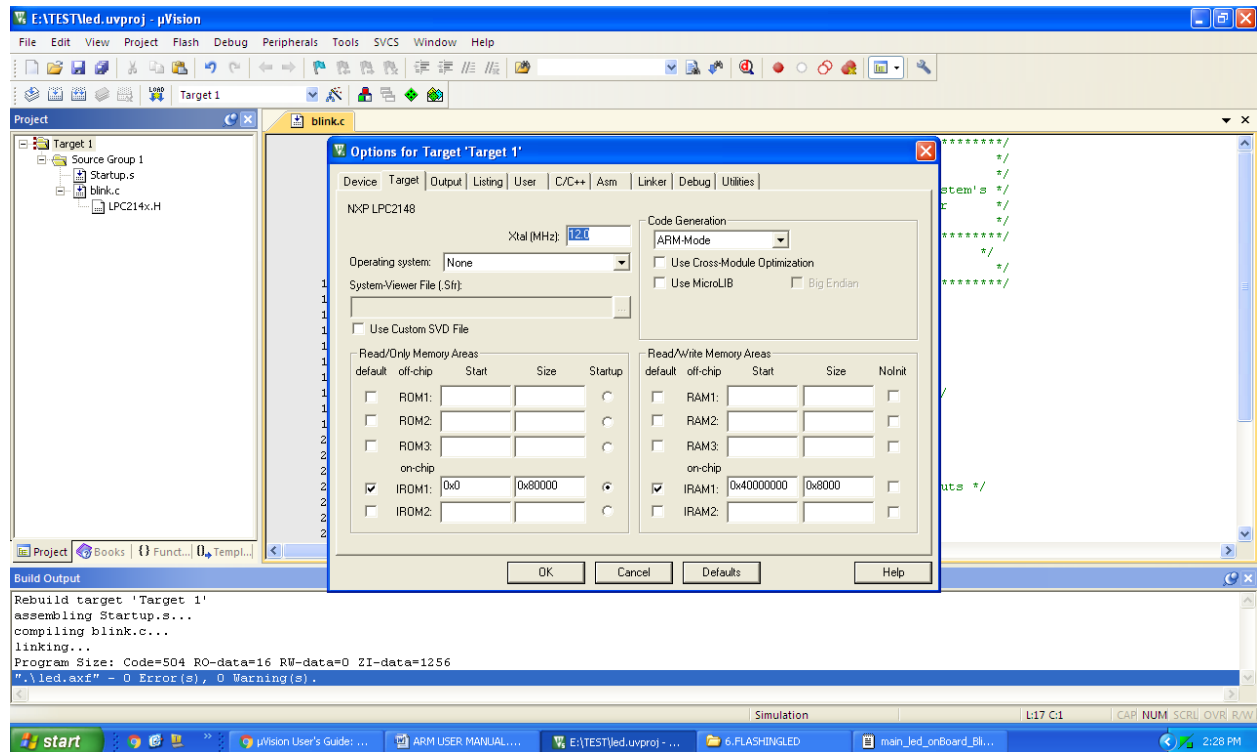
Step28: Again we have goto program window Debug → Start/Stop Debug Session.



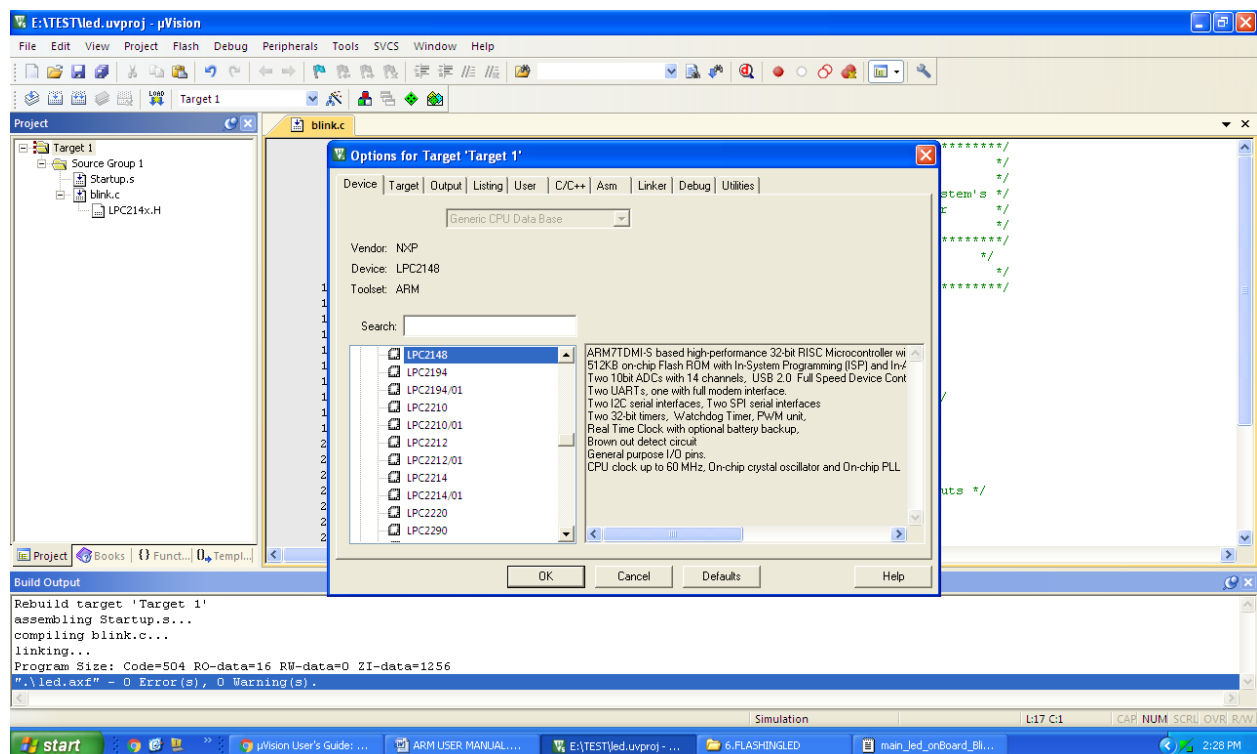
Step29: Create a HEX file. Right click on Target → Options for Target Target1....



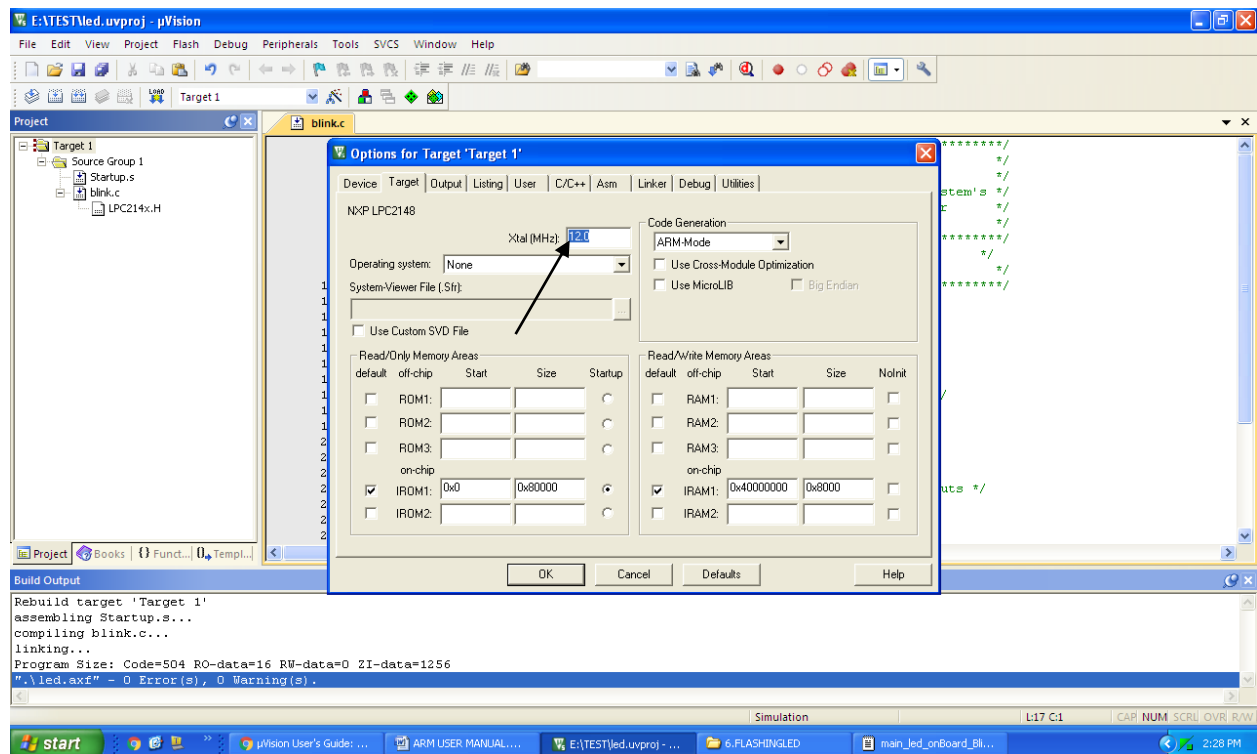
Step30: Target window will open.



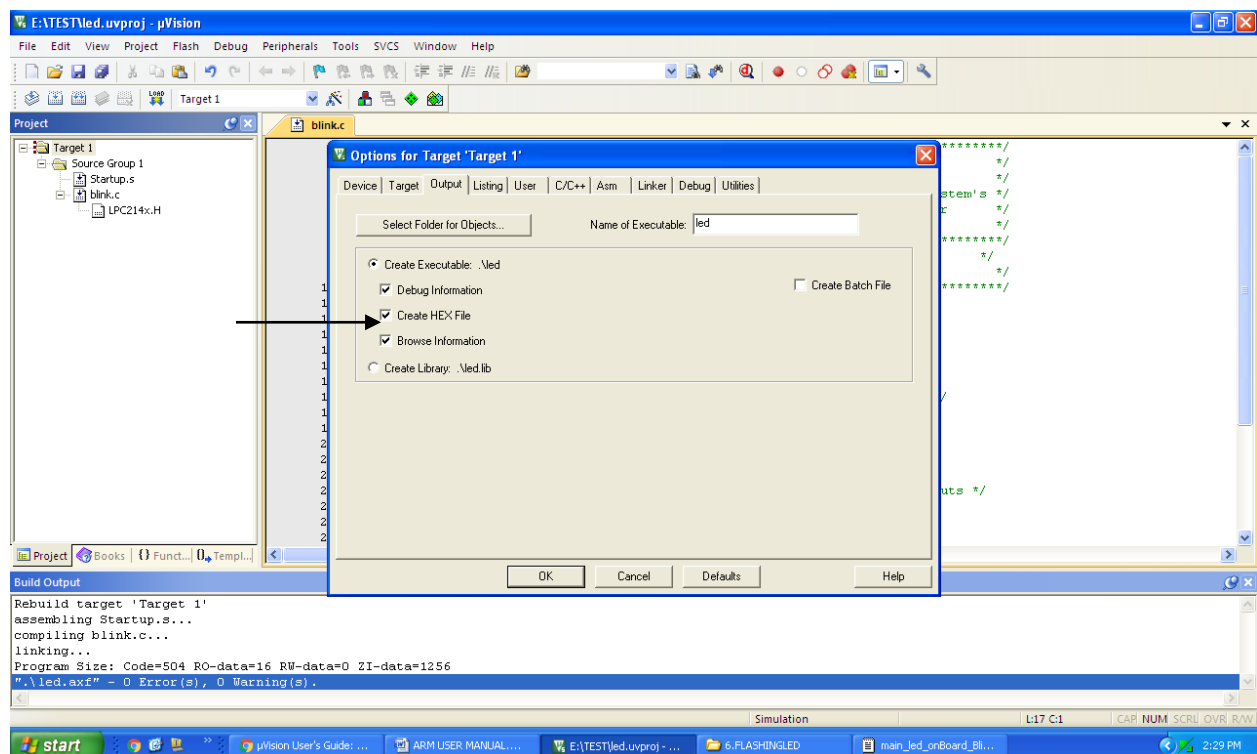
Step31: Select Device → LPC2148.



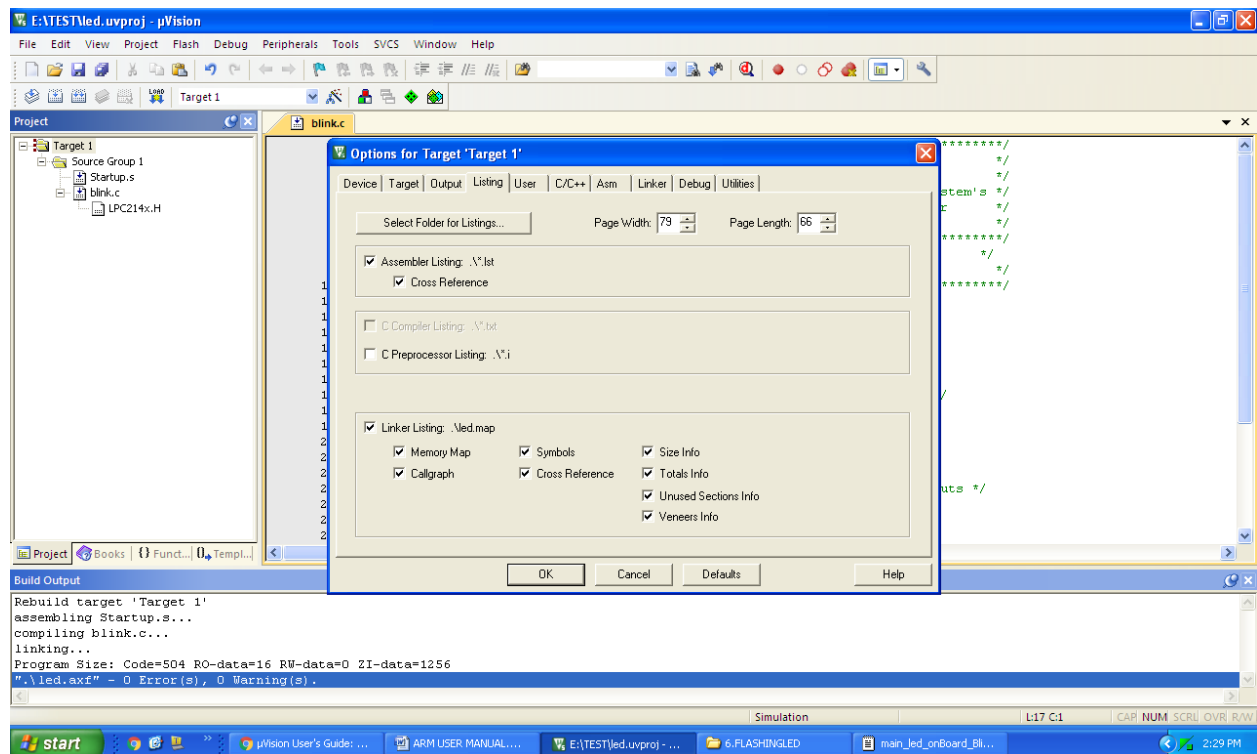
Step32: Select Target and type Xtal [MHz] → 12MHz.



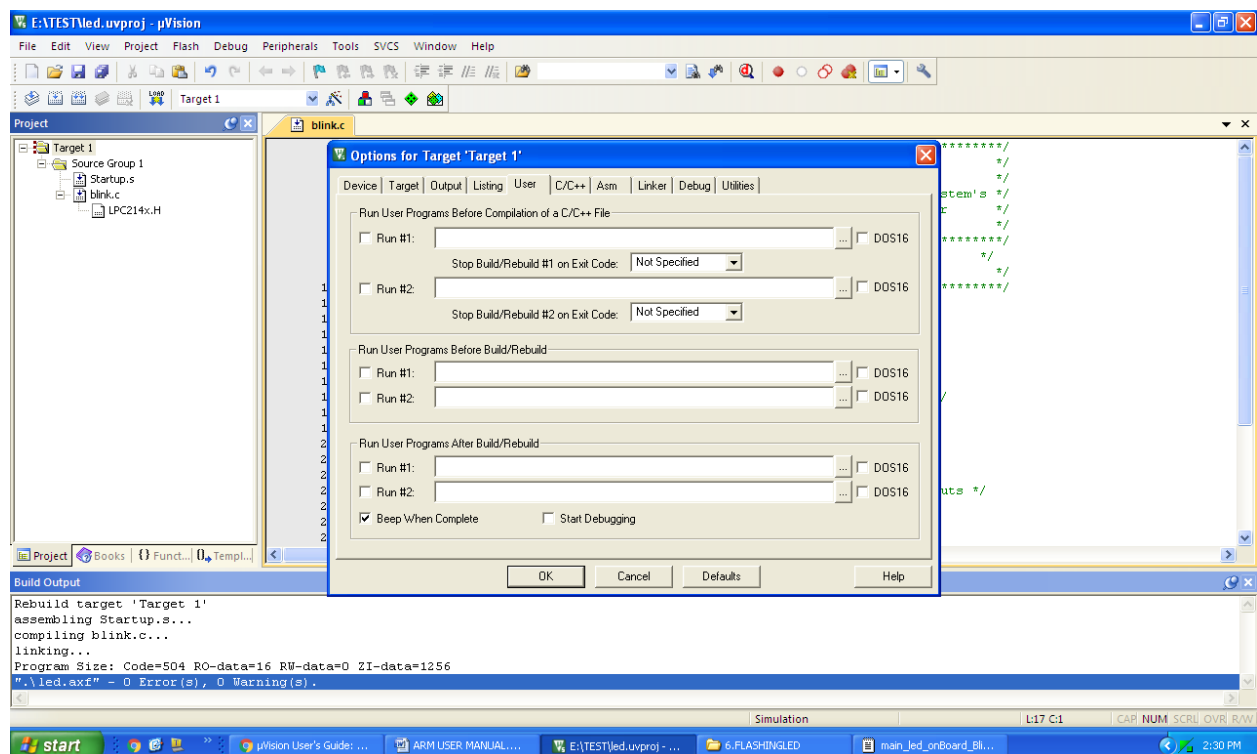
Step33: Select Output → Enable the Create HEX File.



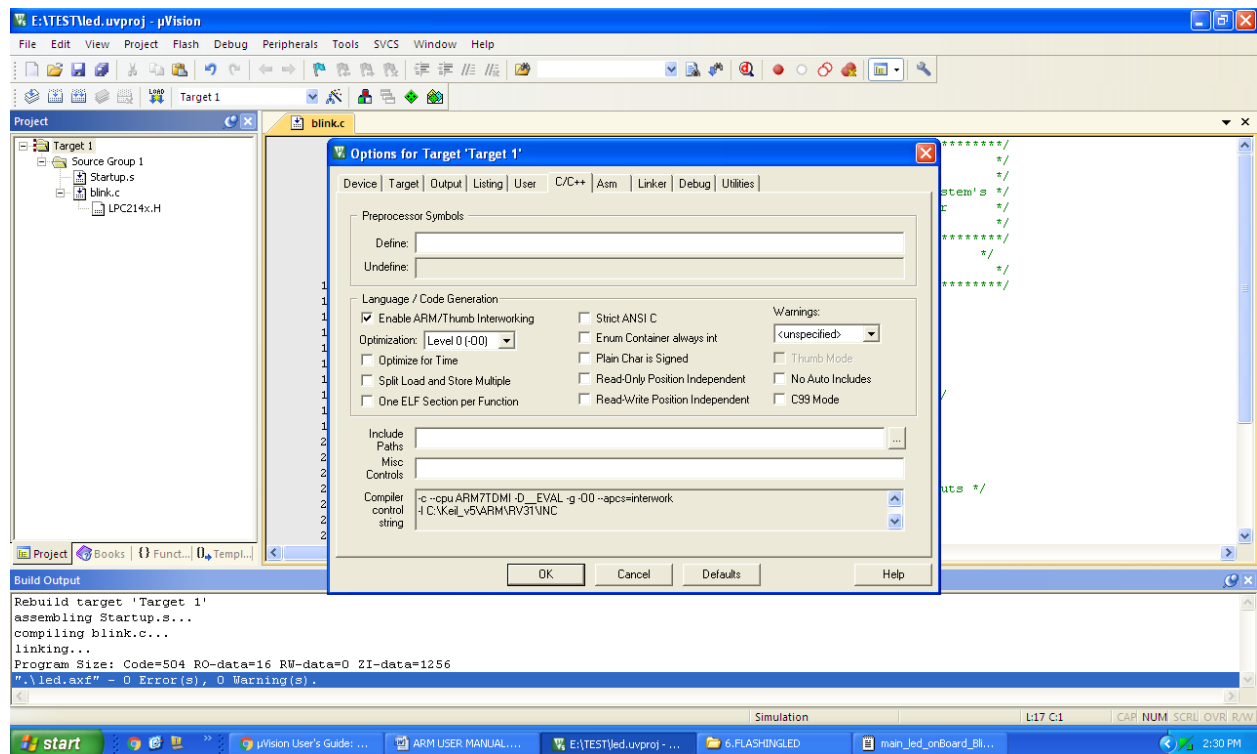
Step34: Select Linking option here No Changes.



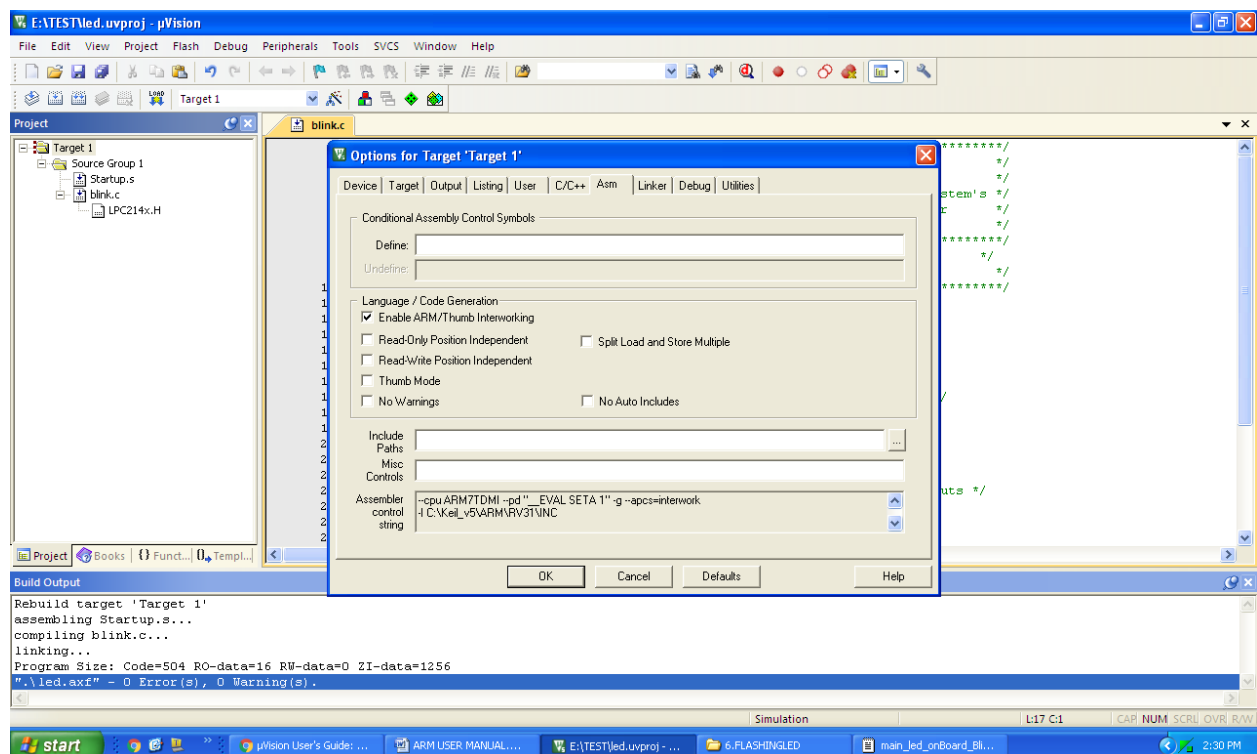
Step35: Select User option here No Changes.



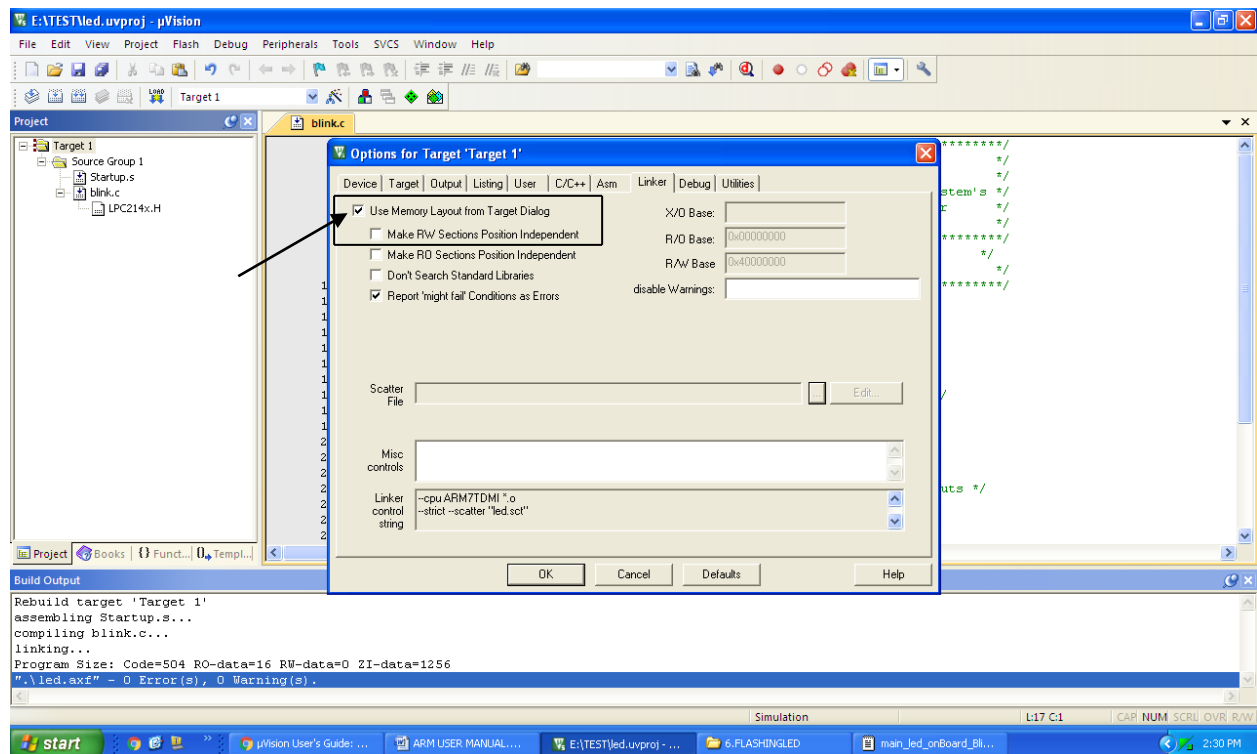
Step36: Select C/C++ option No Changes.



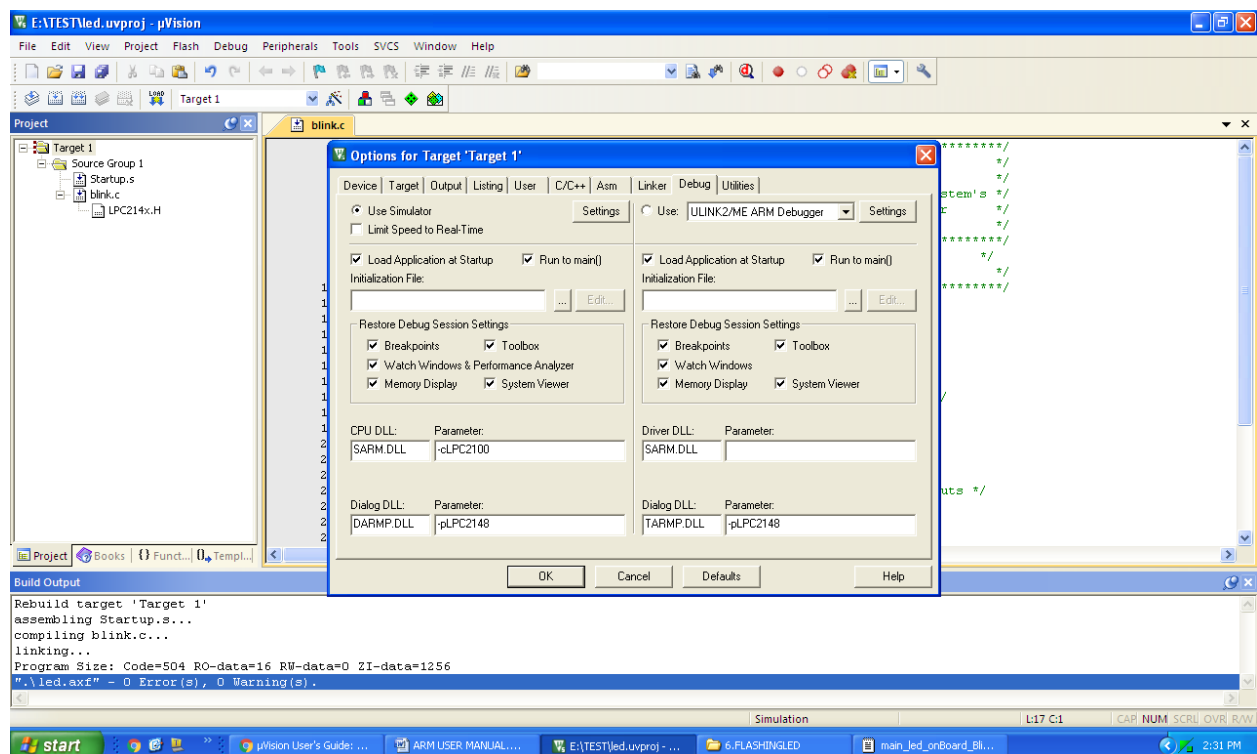
Step37: Select Asm option No Changes.



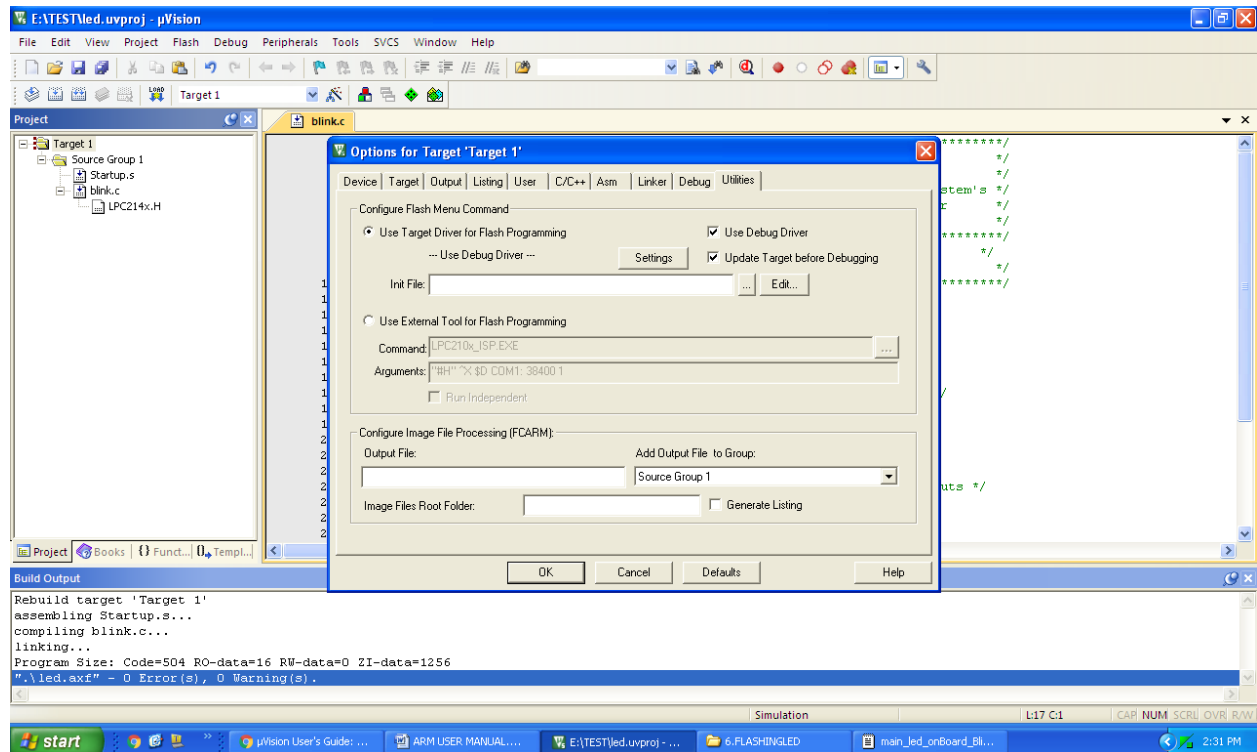
Step38: Select Linker option. Here you have to enable the Target option.



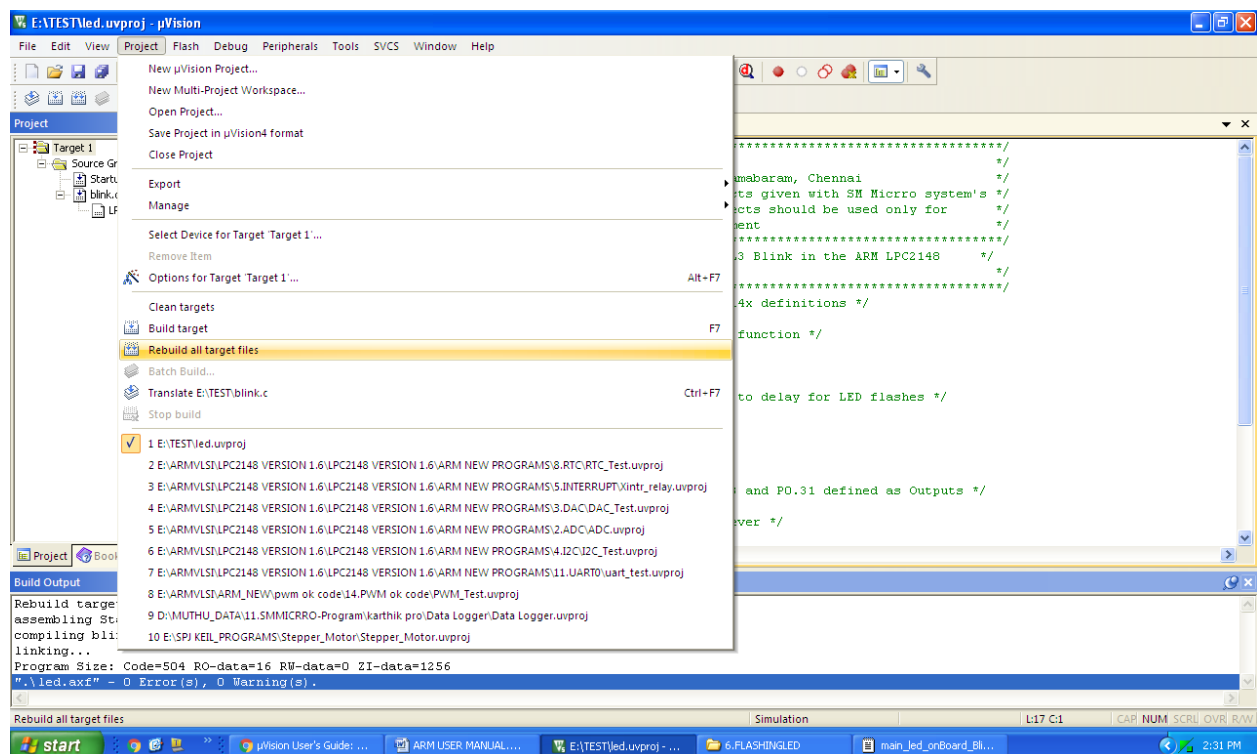
Step39: Select Debug option. Here No Changes.



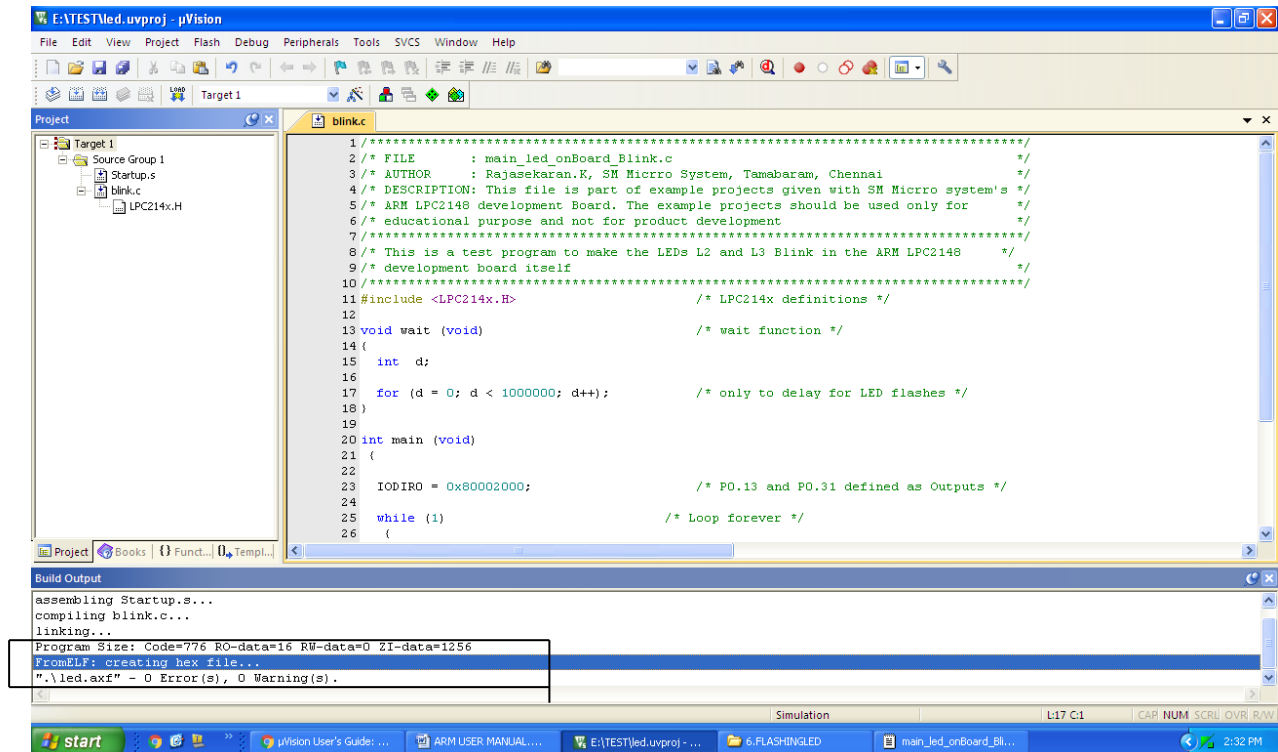
Step40: Select Utilities. Here No Changes.



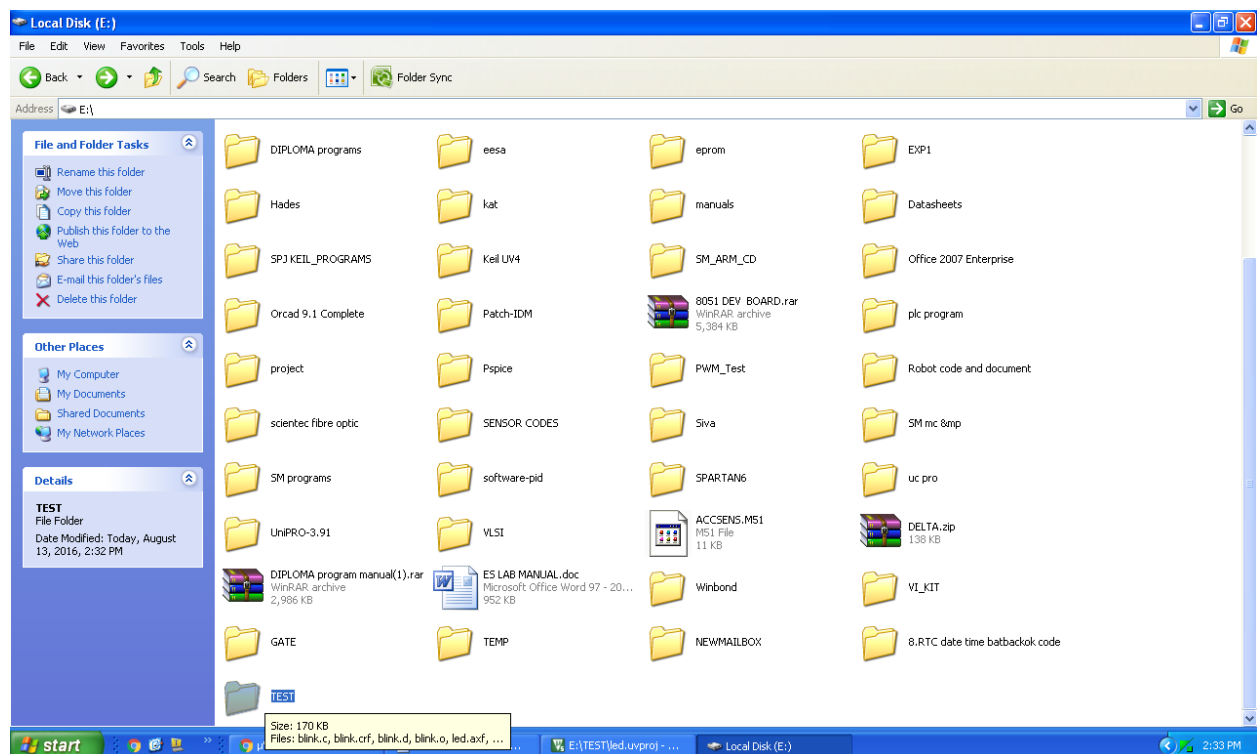
Step41: Next go to Project → Rebuild all target file.

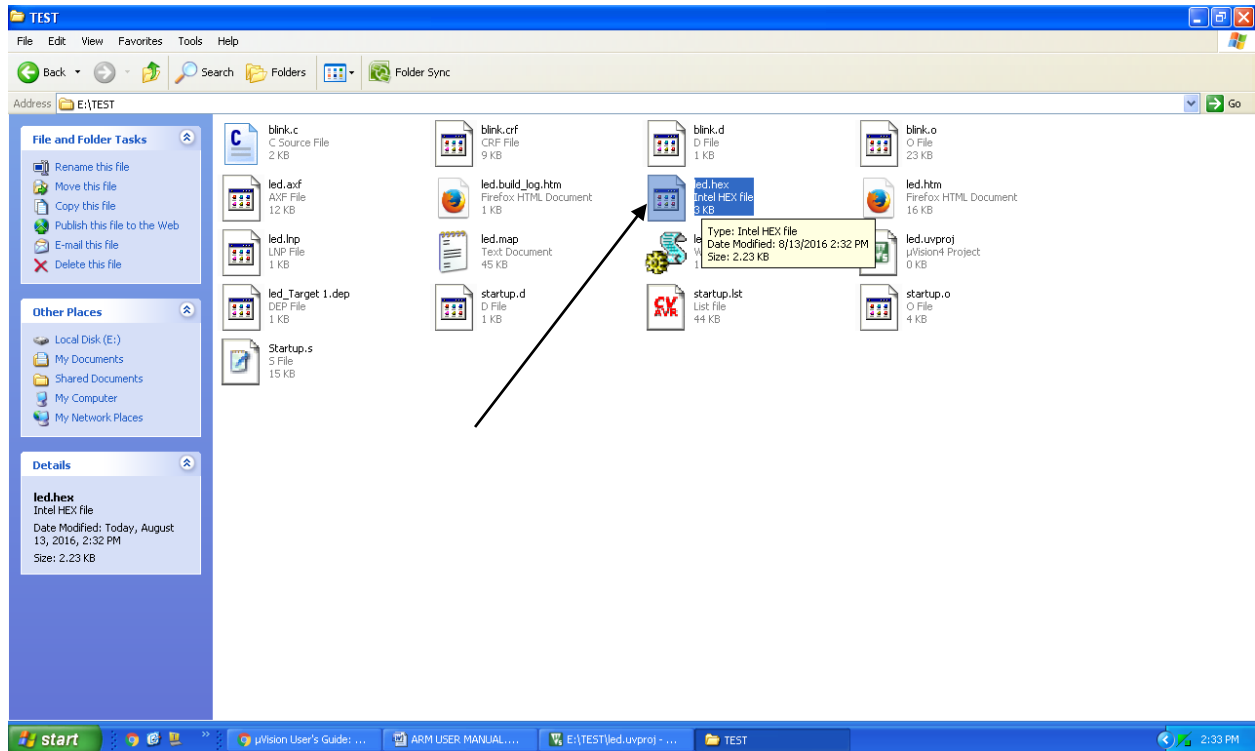


Step42: HEX file was created.



Step43: Check the folder. The HEX file is created or not.

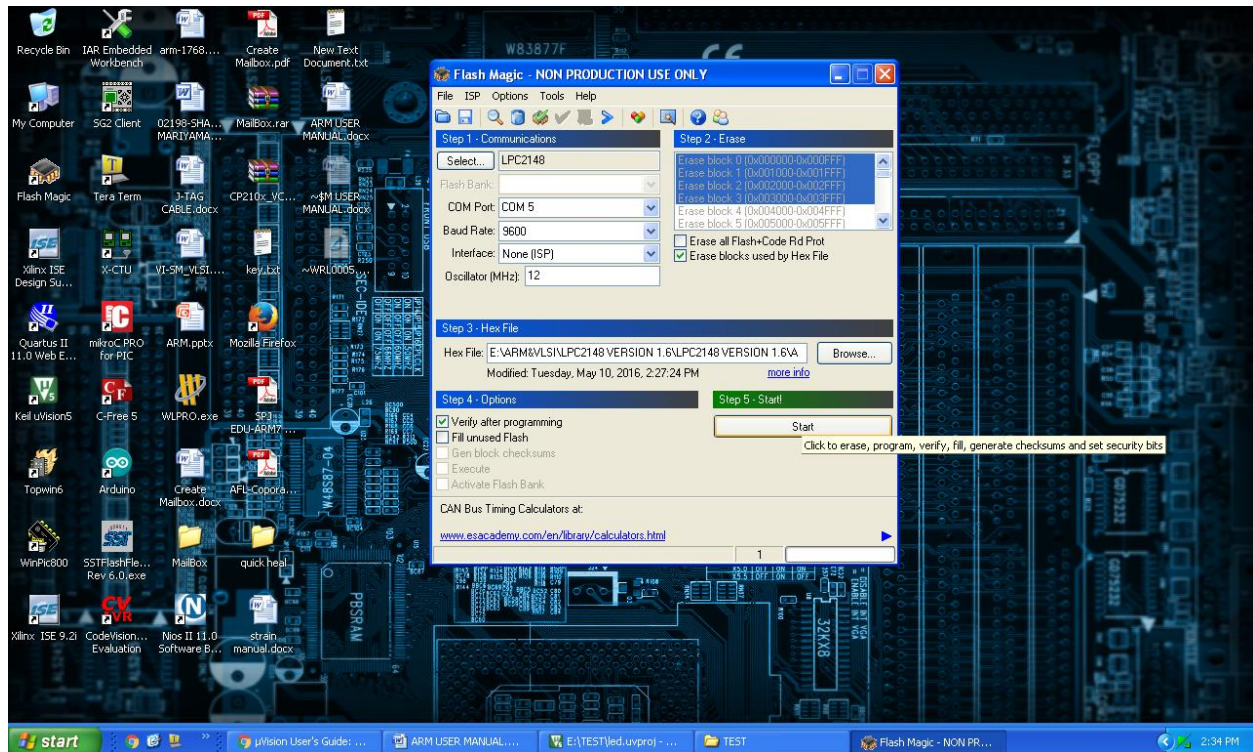




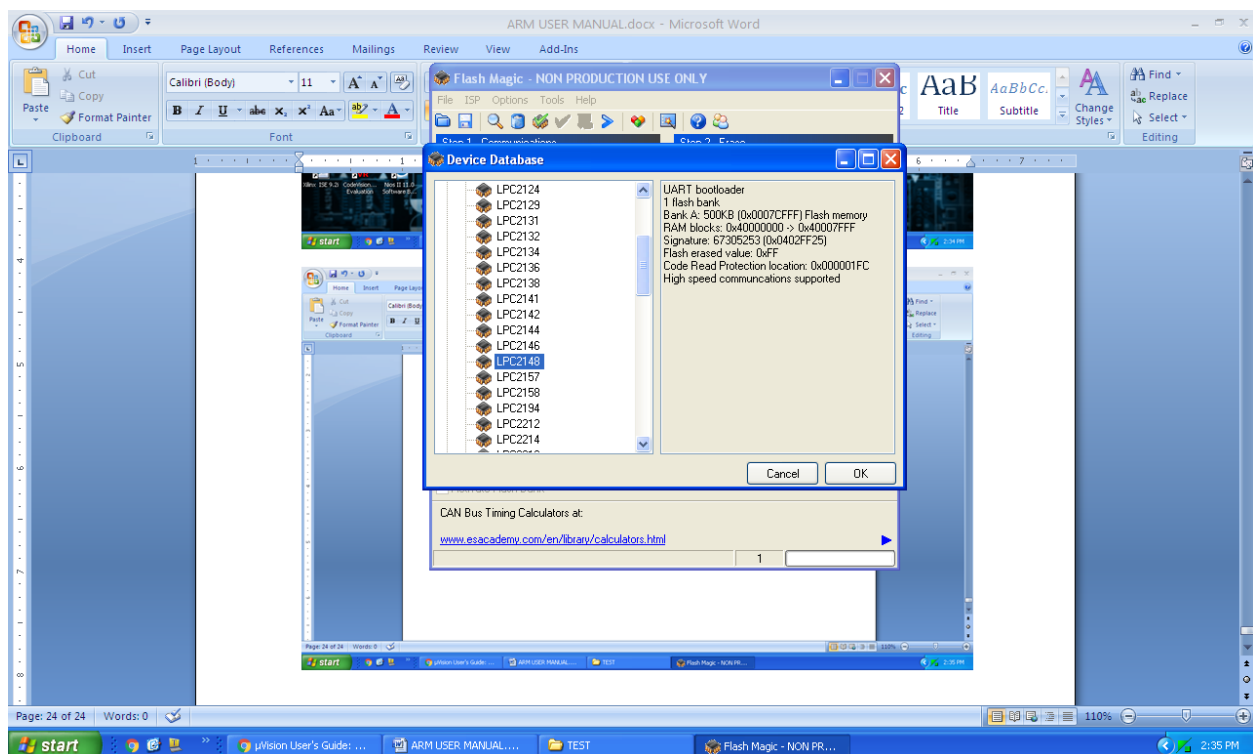
Step44: The HEX file was downloaded in hardware using Flash Magic software.



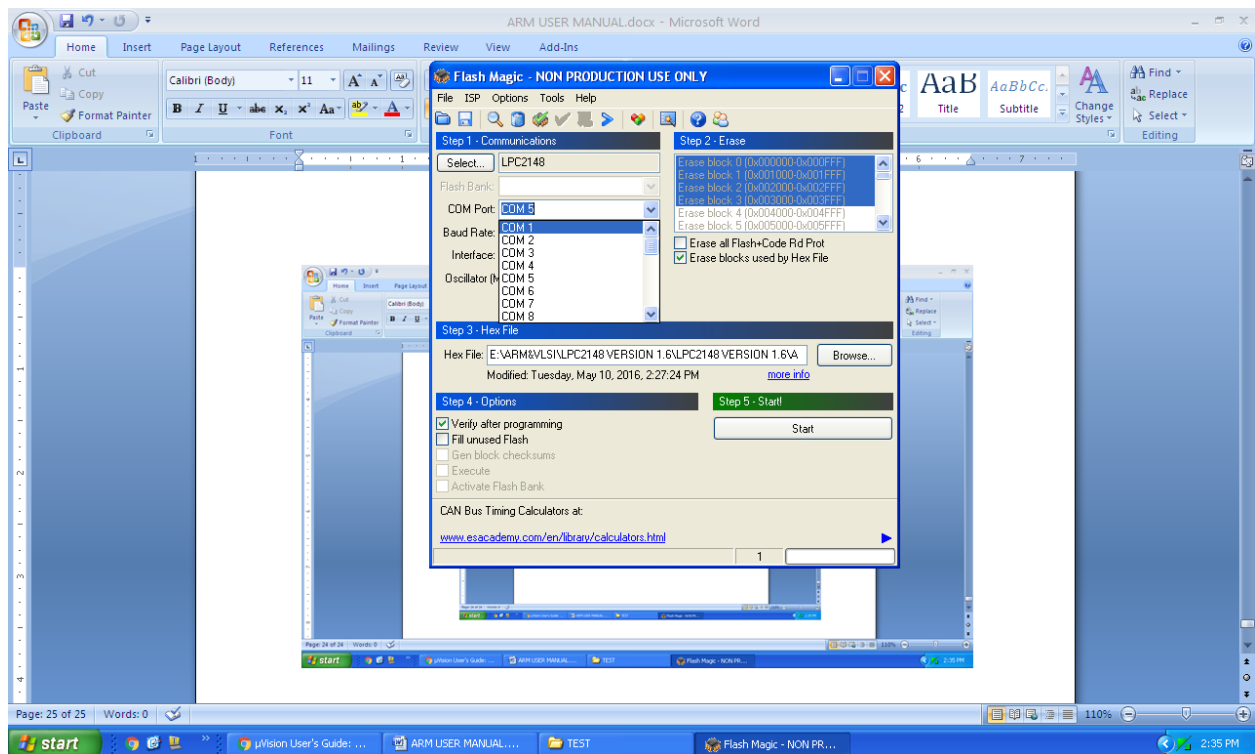
Step45: Open Flash Magic software.



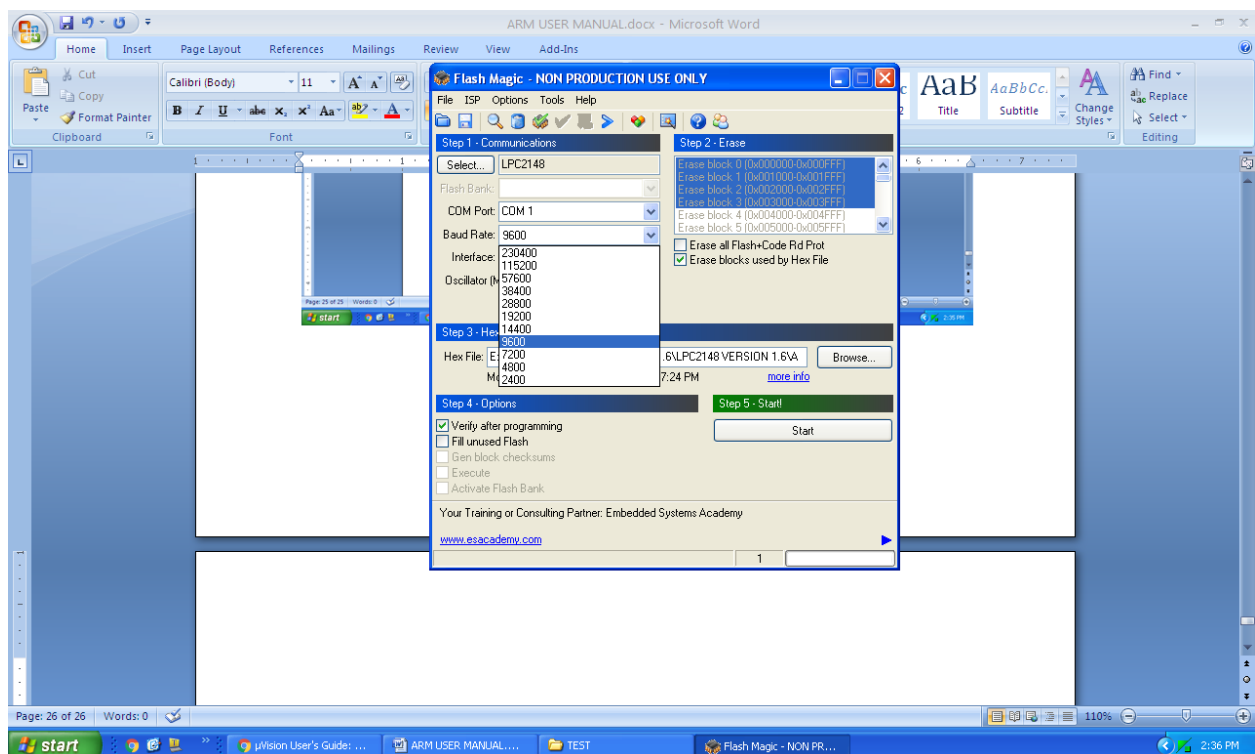
Step46: Select Device LPC2148.



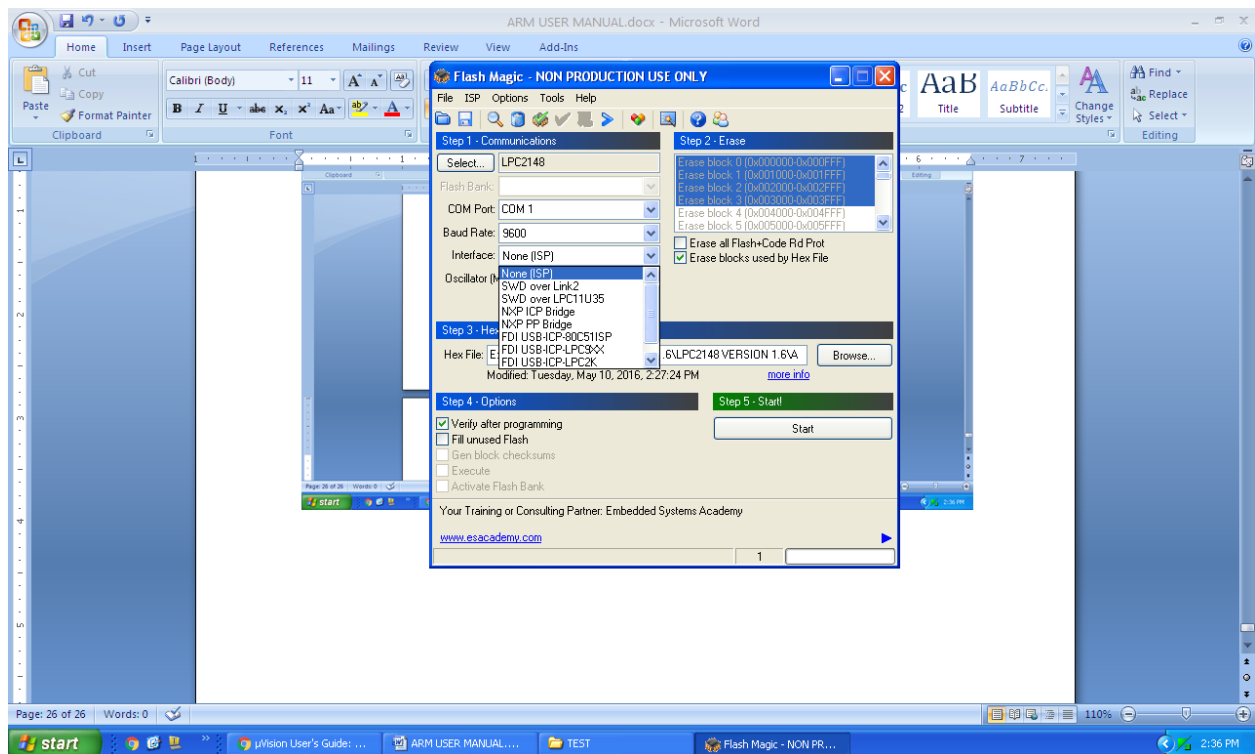
Step47: Select Com Port.



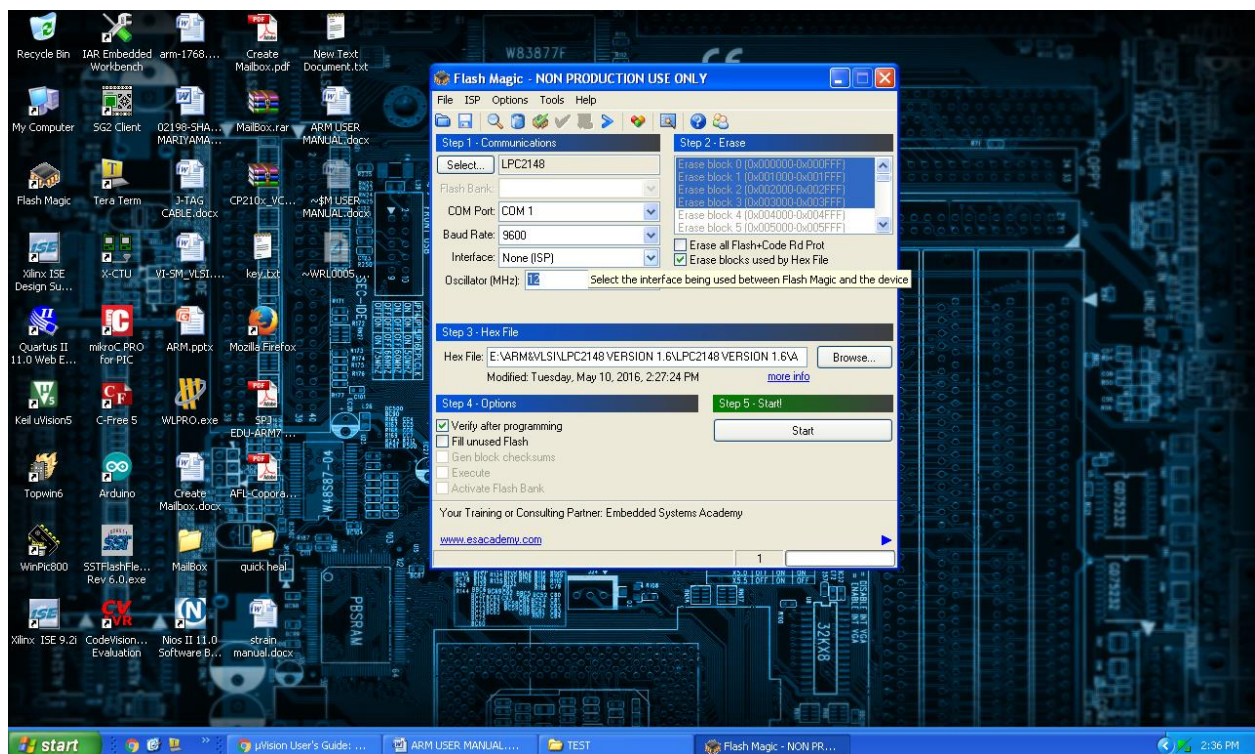
Step48: Select Baud Rate → 9600.



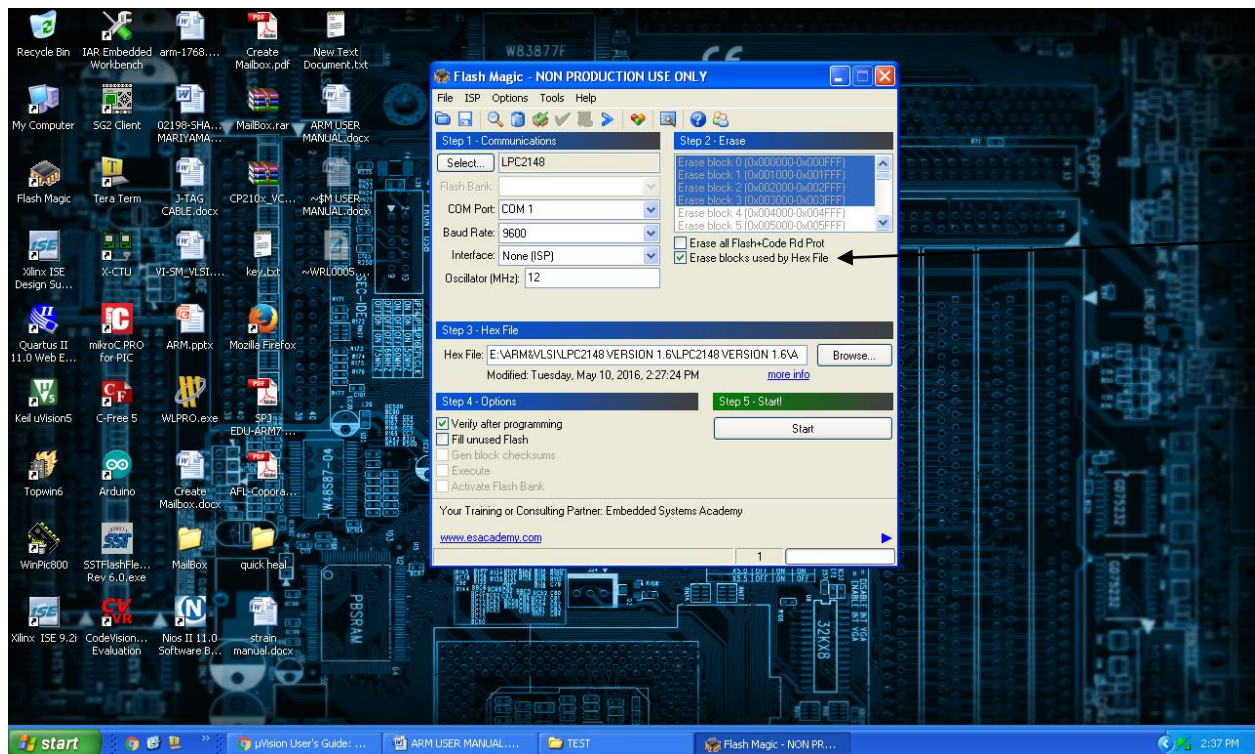
Step49: Select Interface → None (ISP)



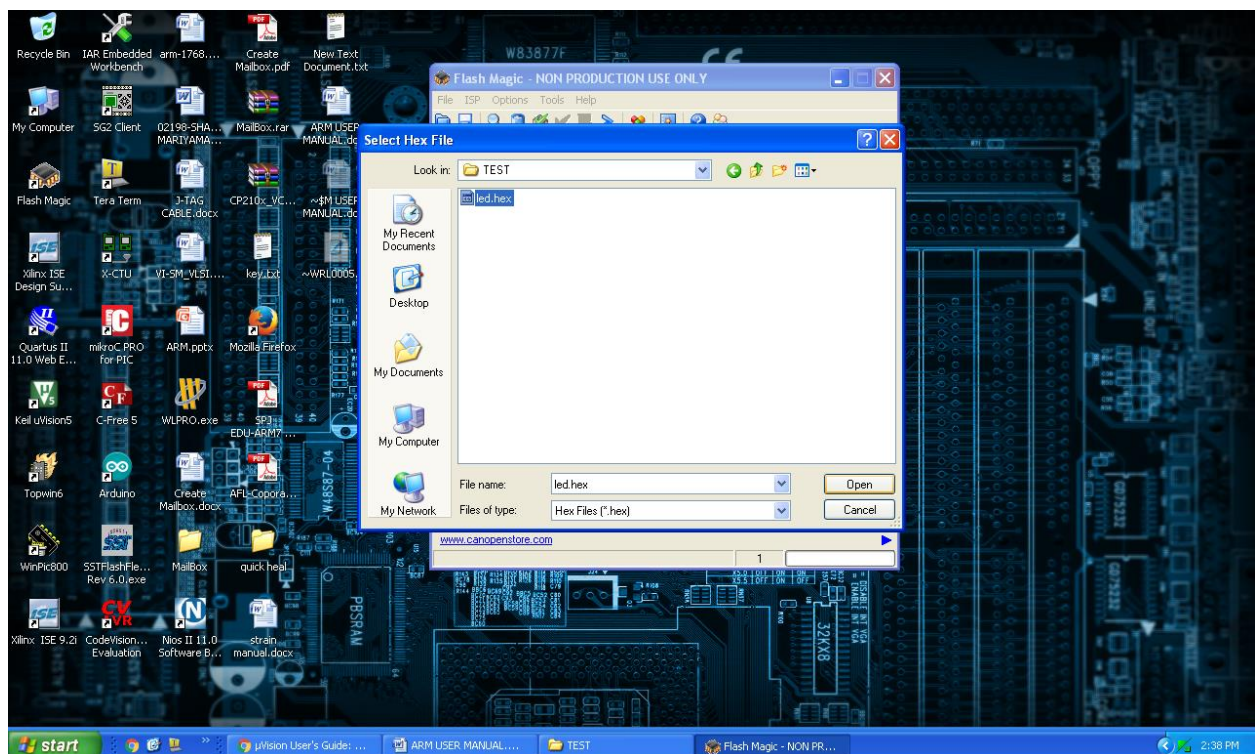
Step50: Oscillator (MHz) → 12.



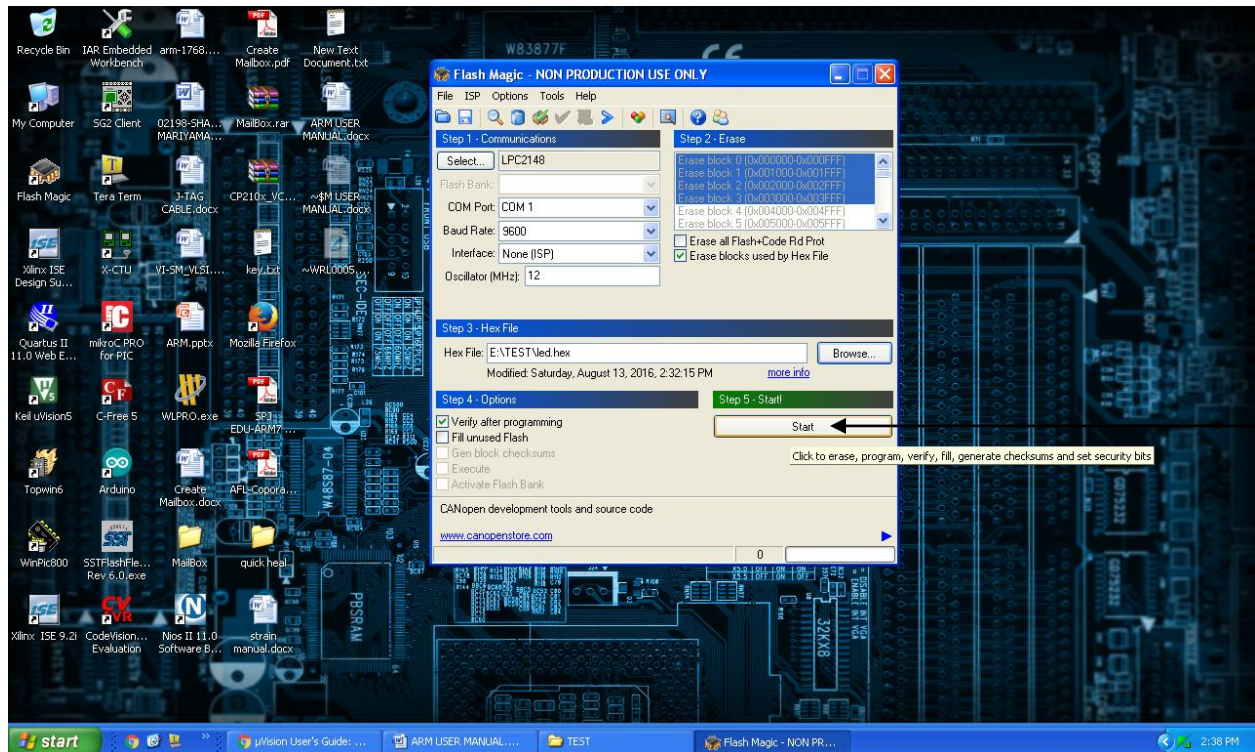
Step51: Select the Erase blocks used by Hex File.



Step52: Click the Browse button and select the hex file.



Step53: Click the Start button.



The file is downloaded and the output will come in hardware.